Labour Supply 1: Individual Attachment to the Labour Market

EC306- Wages and Employment

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Fall 2025

Goals of This Section

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- Introduce common labour market concepts
- Build the labour supply model
- Derive labour supply curve
- Examine effects of labour market policies within that model

Key Labour Force Concepts

Introduction

- The labour market involves several individual decisions about work
  - It is not simply whether to work or not
- In the news you hear about statistics that quantify these decisions
- In this section we cover the most common ones

Labour Force Participation

- The first decision is whether to make yourself *available* for work
- **Labour Force**: people in the eligible population who participate in labour market activities as either employed or unemployed
  - A measure of the people that could work or are working
- Eligible population includes
  - Population 15 years and over
  - Non-institutionalized
  - People not living on reserve
  - Civilians (non-military)

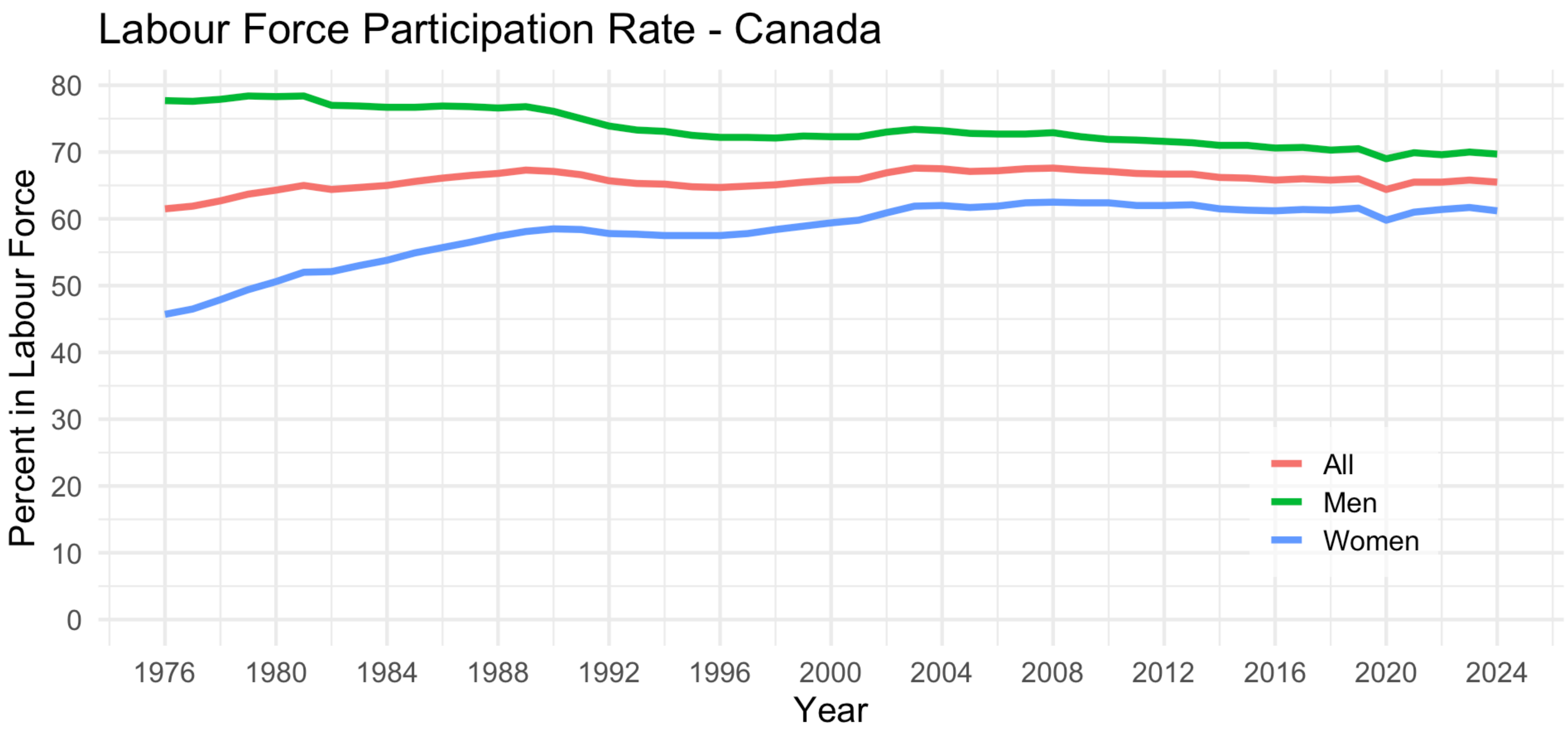
Labour Force Participation

- Eligible people who are not part of the labour force include
  - Retired people
  - People who do only unpaid work in the home
  - Full time students (who are 15 years old and over)
  - People who just give up on looking for work
- **Labour Force Participation Rate (LFPR):** The share of the eligible population (POP) in the labour force (LF)

$$LFPR = LF/POP$$

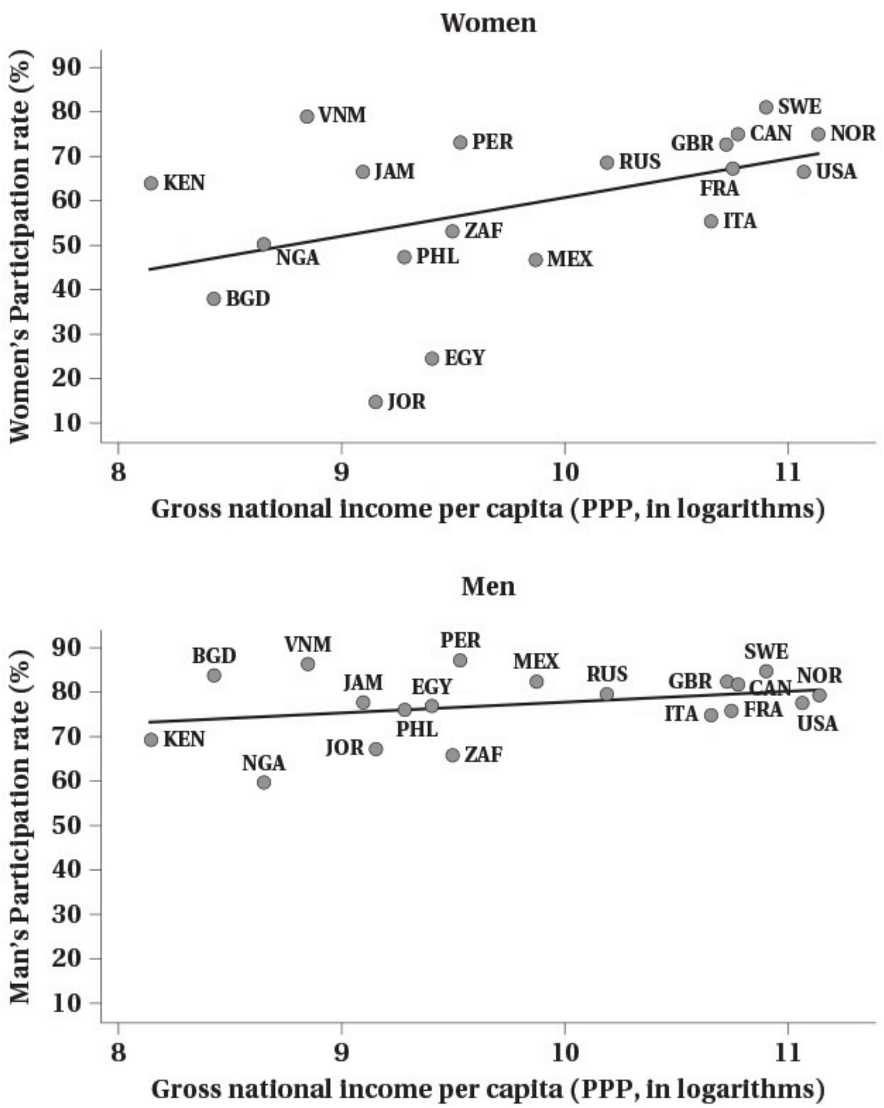
- The LFPR is a common statistic reported in the news

Labour Force Participation



Source: CANSIM Table 14-01-0327-02

Labour Force Participation



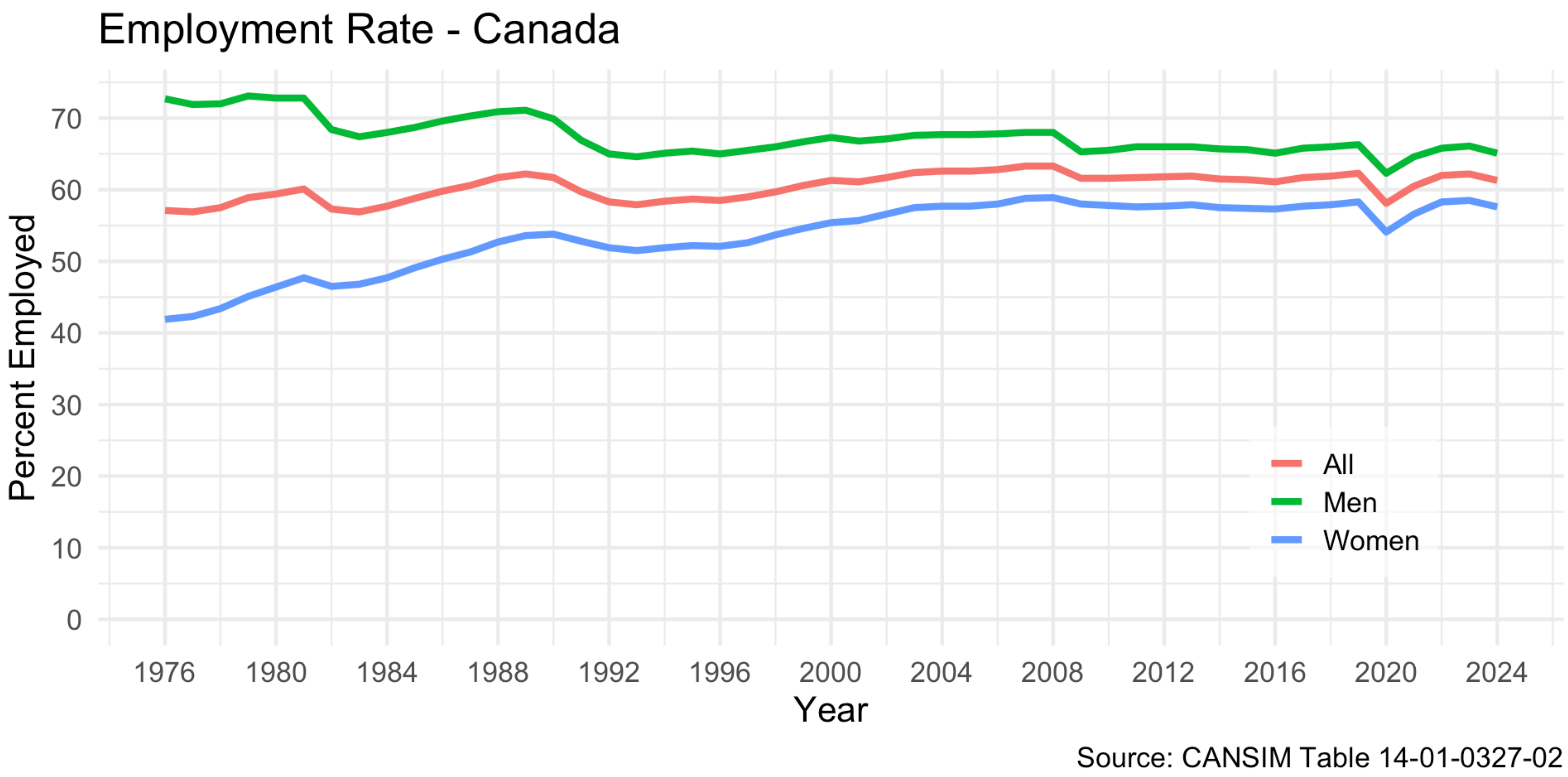
Employment and Unemployment

- People in the labour force are either employed or unemployed
- **Employed (E)**: people who worked during the reference period
  - Includes people normally employed but temporarily absent
- **Unemployed (U)**: people who did not work during the reference period but
  - were available for work and actively looked for work in the past four weeks
  - were waiting to start a new job within the next four weeks
  - on temporary layoff and expecting to be recalled to that job
- The employment rate (ER) and unemployment rate (UR) are

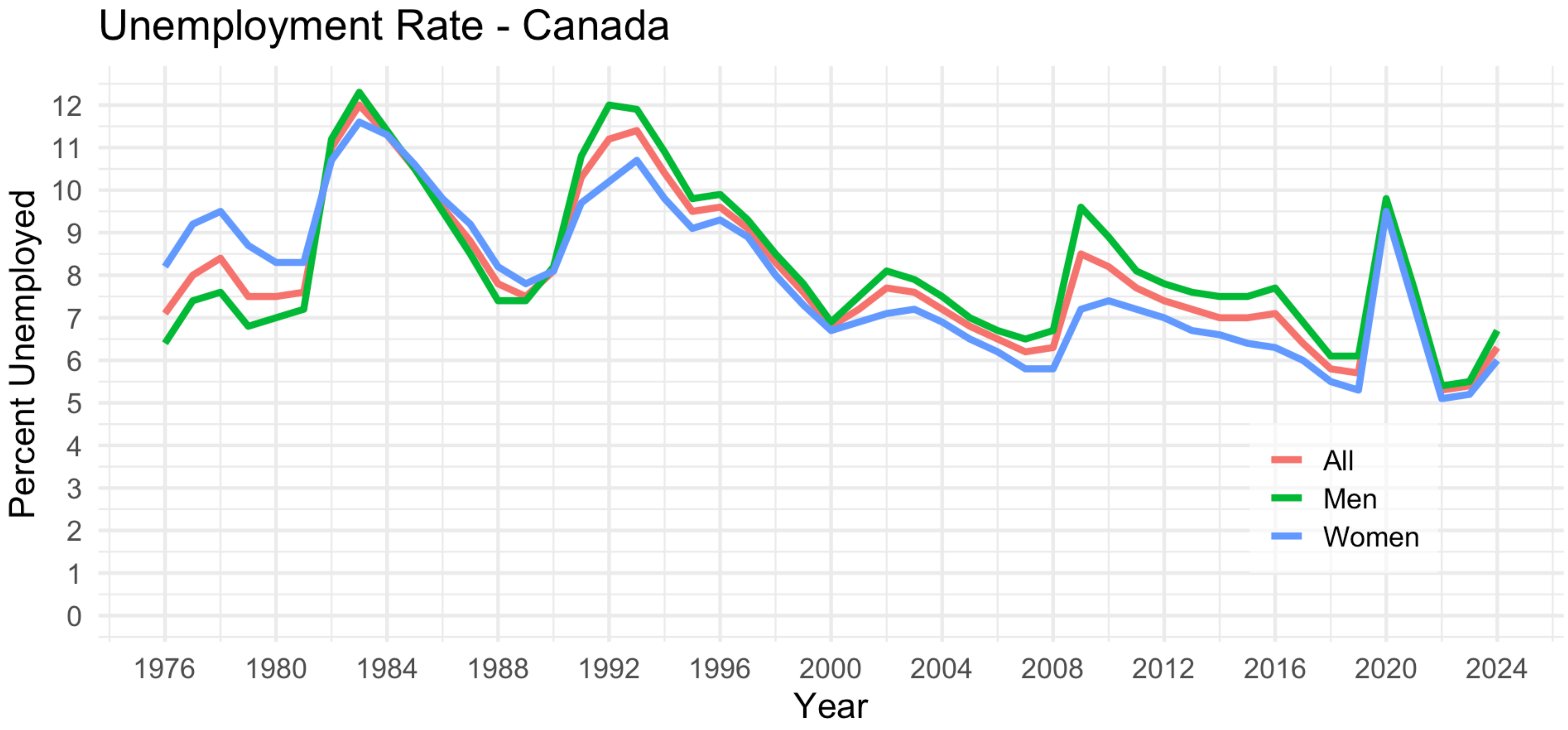
$$ER = E/POP$$

$$UR = U/LF$$

Employment and Unemployment

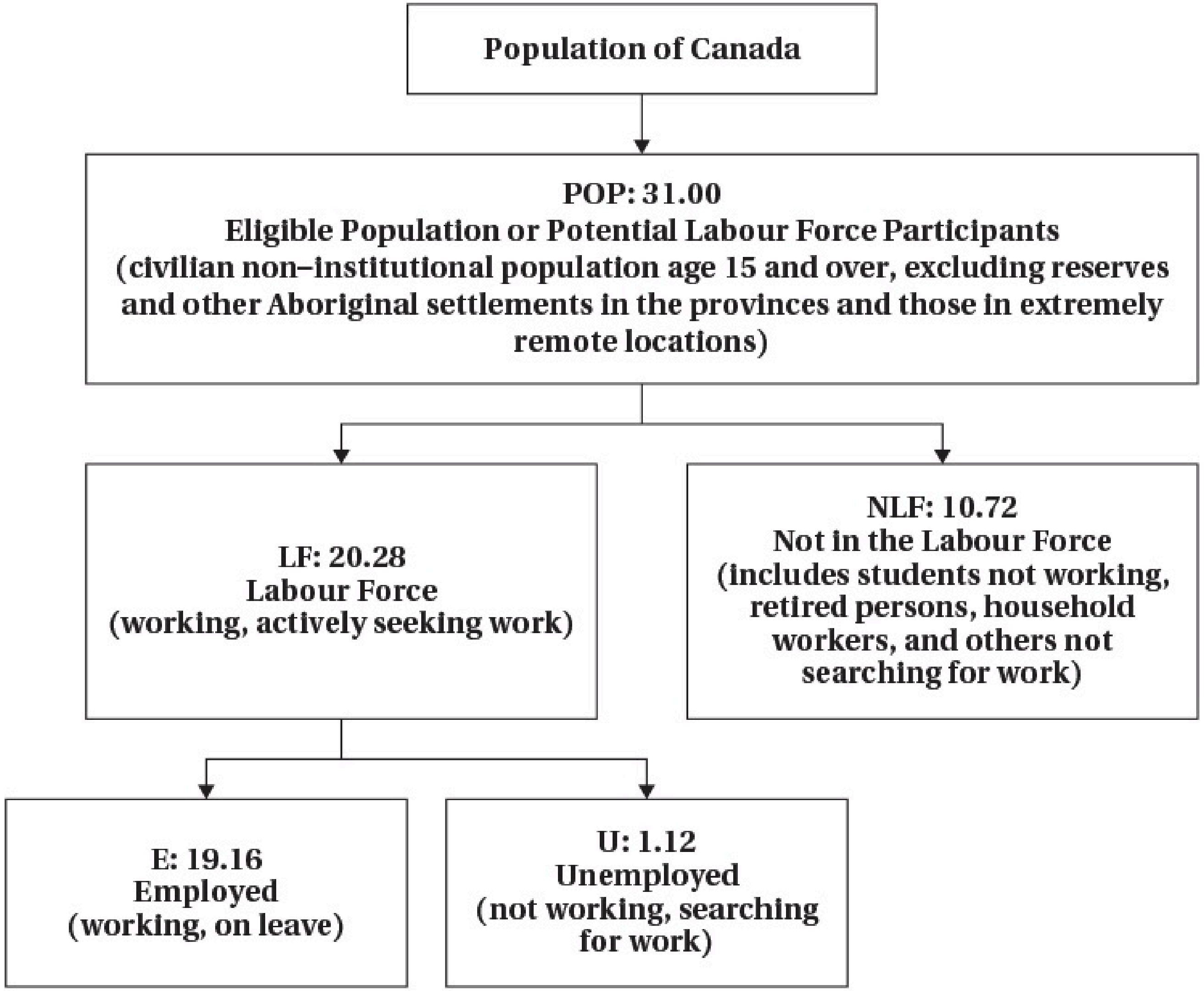


Employment and Unemployment



Source: CANSIM Table 14-01-0327-02

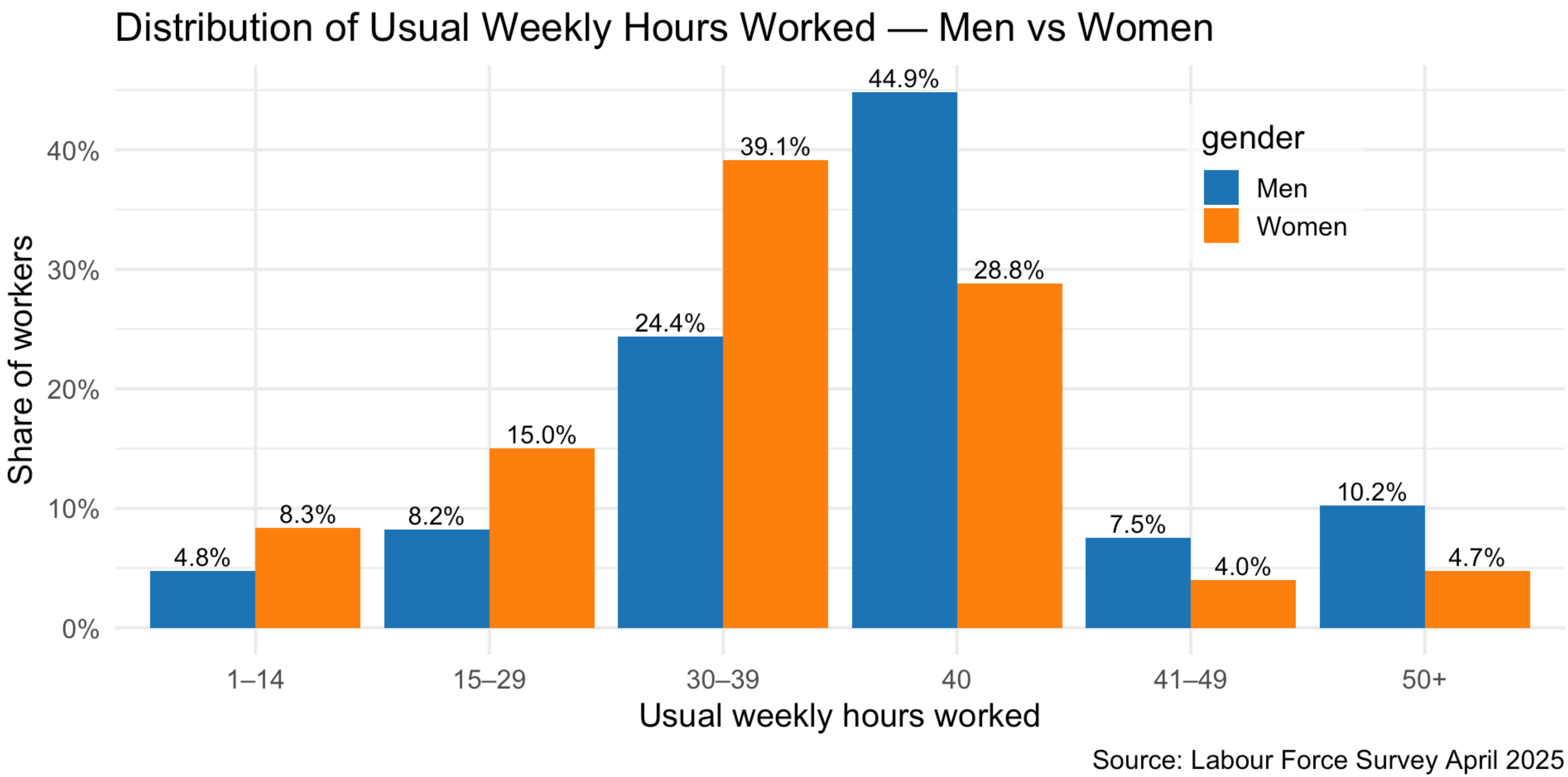
Labour Force Hierarchy



Hours Worked

- Individuals decide not only whether to work or not, but how intensively
- Some people work full time, some work part time
  - Work arrangements can be flexible
  - “Gig” jobs and remote work have increased that flexibility
- Recent data show
  - Almost half of men work full time hours
  - About 1/4 of women work full time hours

Hours Worked



Labour Supply Model

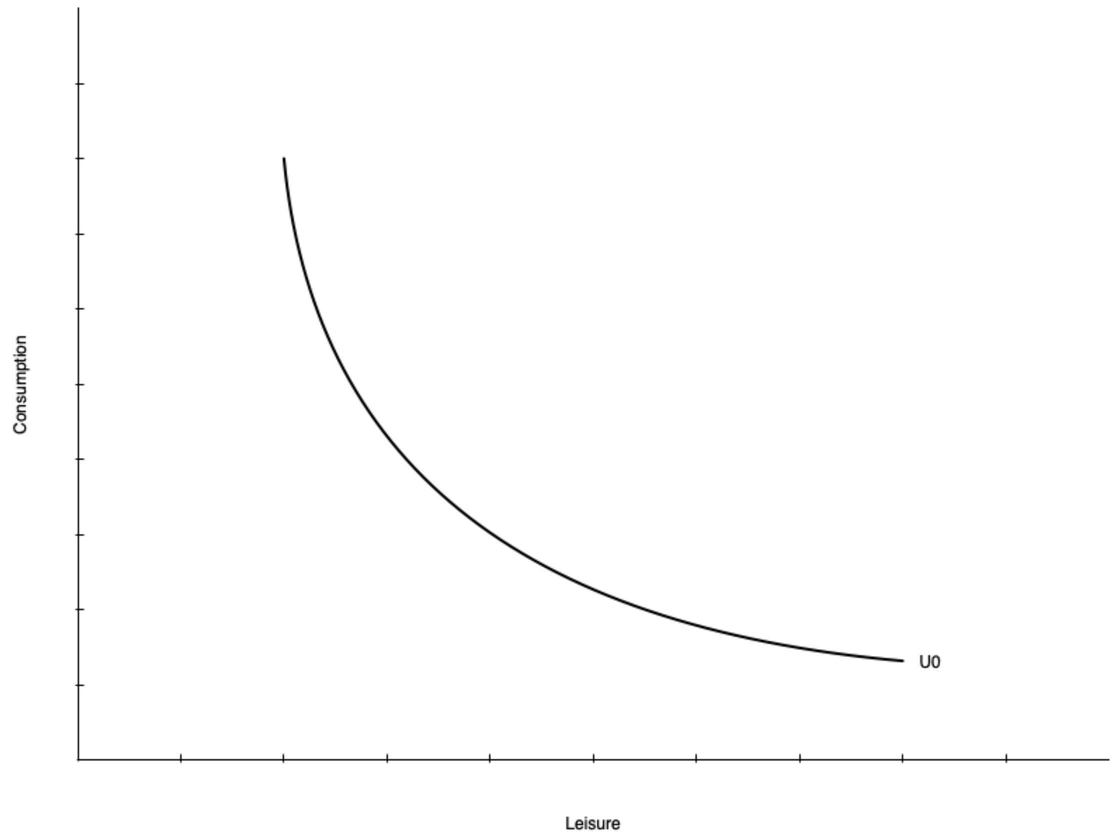
Introduction

- In this section we introduce a model we use throughout the course
- The model helps us understand how individuals make decisions about work
- Based on utility maximization as you saw in EC270
- Individual choose between two goods
  - Consumption of goods and services
  - Leisure
- Choices are constrained by time, wage rate, and non-labour income
- There are a few wrinkles that make this market unique

Preferences

- Individuals have preferences over consumption (C) and leisure (L)
  - People like consuming goods and services
  - Also like having non-work time
    - Includes fun activities but also non-work tasks like chores, education, etc.
- The tradeoffs people are willing to make between C and L are represented by indifference curves
  - At every point along the curve the individual is equally happy
  - Curves further from the origin represent higher levels of utility

Preferences

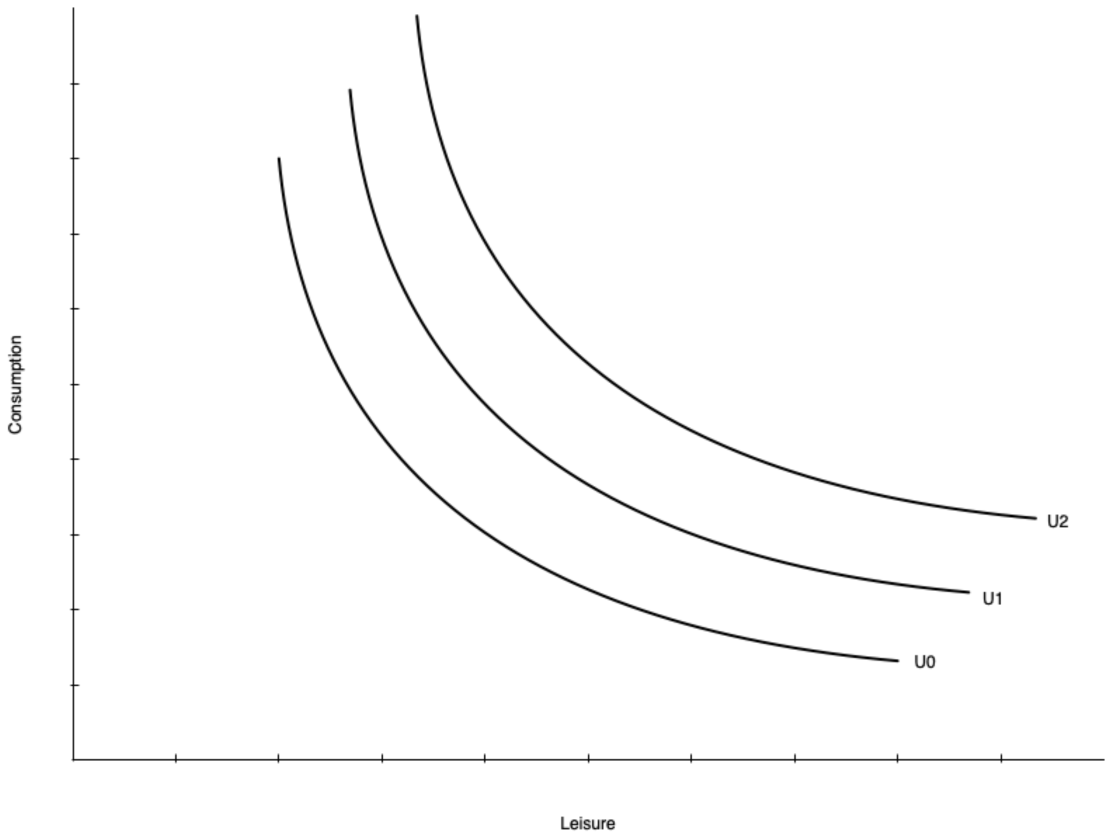


- On the left is a single indifference curve
- Any point represents equal utility (U_0)
- **Marginal Rate of Substitution (MRS)**: the rate at which a person is willing to trade consumption for leisure while remaining equally happy
- The MRS is the slope of the indifference curve at a point
- Individuals have diminishing MRS
  - Willing to give up less consumption for additional leisure the more leisure they already have
  - Slope gets shallower with more leisure

Preferences

- Indifference curves do not have to be smooth curves like the one shown
- Could be straight lines
  - Means the tradeoff between consumption and leisure is constant
  - Willingness to trade away leisure for consumption does not depend on leisure
- Could be “L” shaped
  - Consumption and leisure must be consumed in fixed proportions
- Mostly we will deal with smooth preferences

Preferences

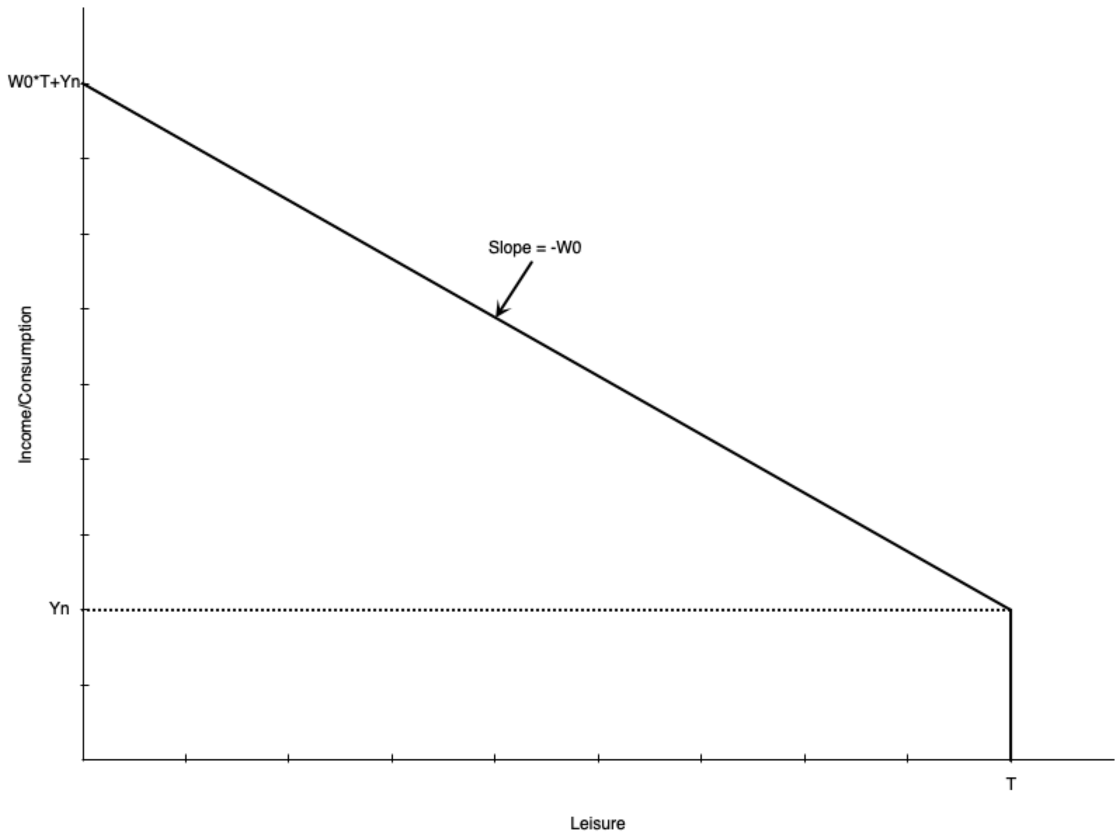


- Utility increases with curves further from the origin
  - $U_2 > U_1 > U_0$

Constraints

- Individuals do not choose consumption and leisure bundles on preference alone
- They are constrained by their budget
- The budget is determined by
  - The wage rate
  - Total number of hours available
  - Non-labour income
- Budget constraint is also called
  - Potential income constraint
  - Full income constraint

Constraints



- Graph shows a linear budget constraint
- Assume no saving, so consumption = income
- There are T total hours to allocate to work or leisure
- Individual earns Y_n non-labour income
  - Earned even with zero work hours (full leisure hours)
- Each hour worked gets wage of w
- Person working T hours (no leisure) earns $WT + Y_n$ income
  - Called “full income”

Constraints

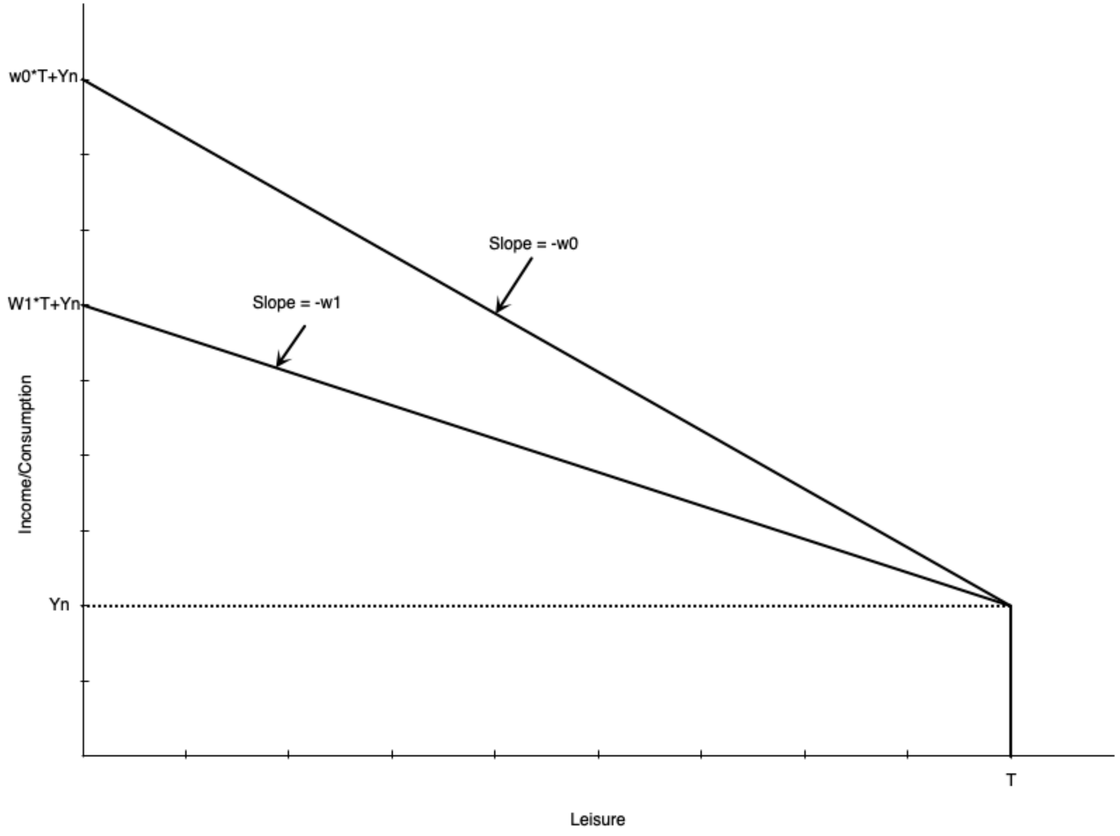
- Mathematically, the budget constraint is

$$C = w(T - L) + Y_n$$

- Consumption is equal to income from working $(w(T - L))$ plus non-labour income (Y_n)

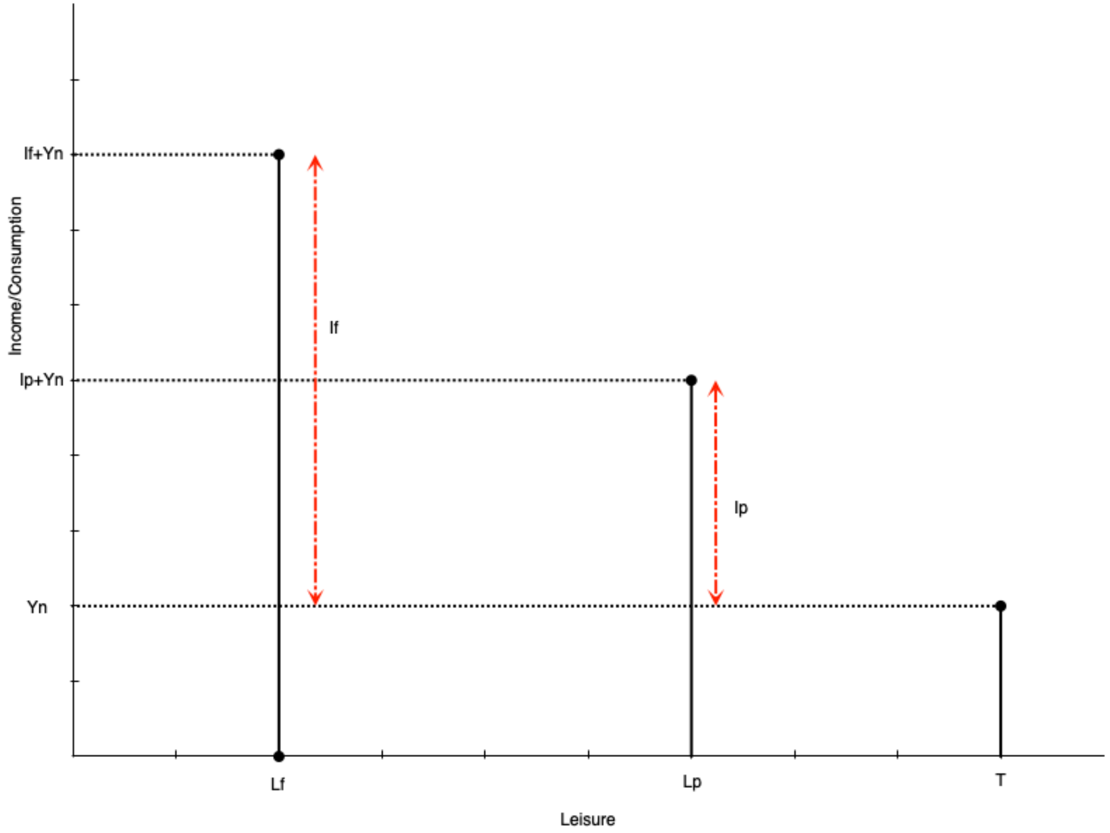
- The slope of the budget constraint is $-w$
  - The opportunity cost of leisure is the wage rate
  - Each hour of leisure means one less hour worked, which means w less income
- Someone who works zero hours has T leisure hours and Y_n consumption
- Someone who works T hours has zero leisure hours and $wT + Y_n$ consumption

Constraints



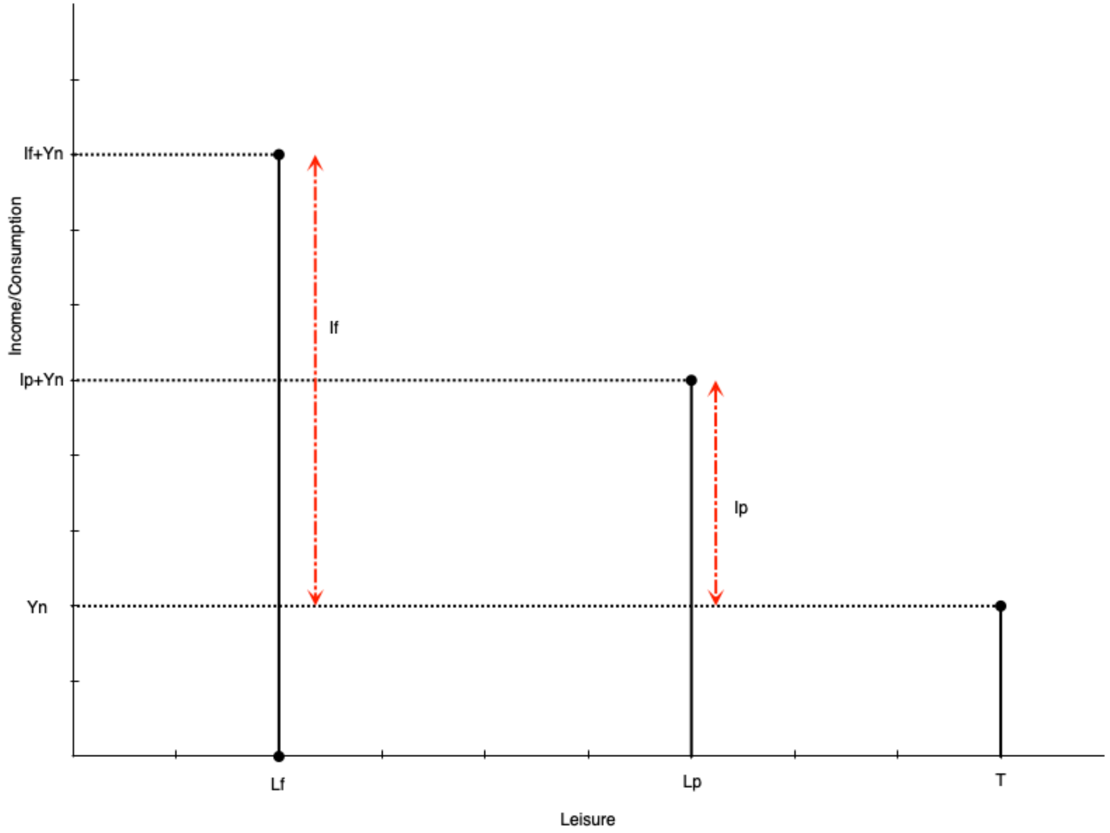
- Slope is the negative of the wage rate w
- A lower wage means shallower slope
  - If $w_1 < w_0$, the budget line swivels down
  - They intersect at leisure = T because there are no work hours
- Lower wage also means full income is lower
  - Working all hours leads to less total labour earnings

Constraints



- Often people cannot adjust their work hours freely
- They make only be able to work full time or part time
- Budget constraint is no longer a line, but a set of points
- Suppose an individual chooses part time work
  - Work hours are $h_p = T - L_p$
  - Labour earnings are $I_p = w * h_p$
  - Total consumption (and income) is $C_p = I_p + Y_n$

Constraints

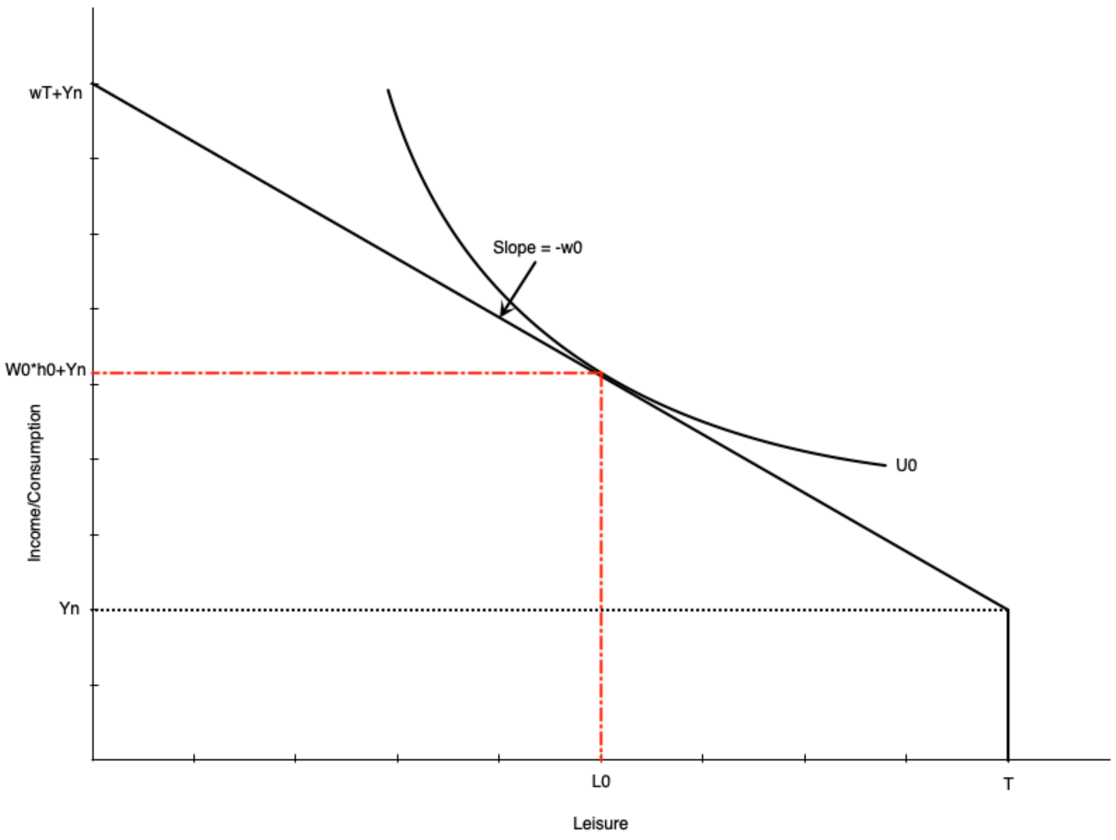


- A person working full time hours
  - Work hours are $h_f = T - L_f$
  - Labour earnings are $I_f = w * h_f$
  - Total consumption (and income) is $C_f = I_f + Y_n$
- In theory wage could also be different for full time work

Consumer Optimum

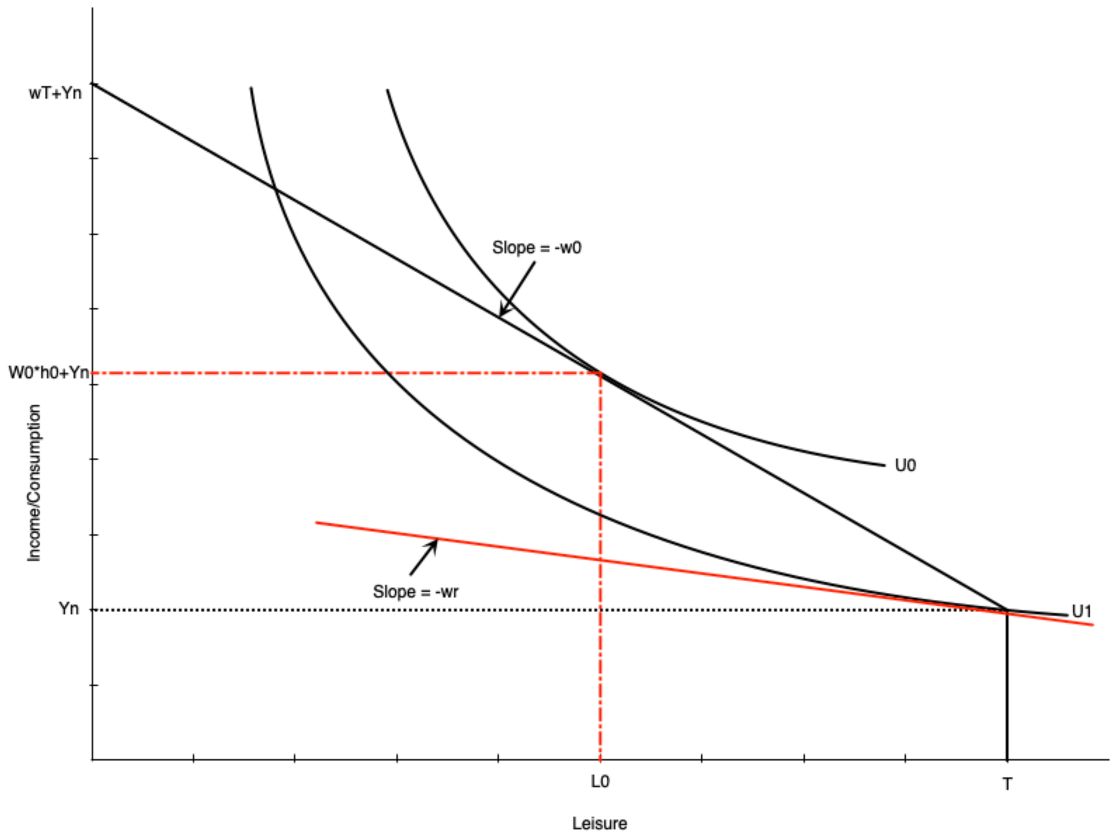
- Work hours are chosen using the combination of preferences and constraints
- The consumer optimum is the point that maximizes utility subject to the budget constraint
- The optimum occurs where the budget constraint is tangent to an indifference curve
- There are two possibilities for the optimum
  - Non-participation: optimum occurs at the end of the budget constraint (full leisure)
  - Participation: optimum occurs at an interior point (some leisure, some work)
- Model also defines the **reservation wage**
  - Lowest wage rate at which a person will choose to work

Consumer Optimum



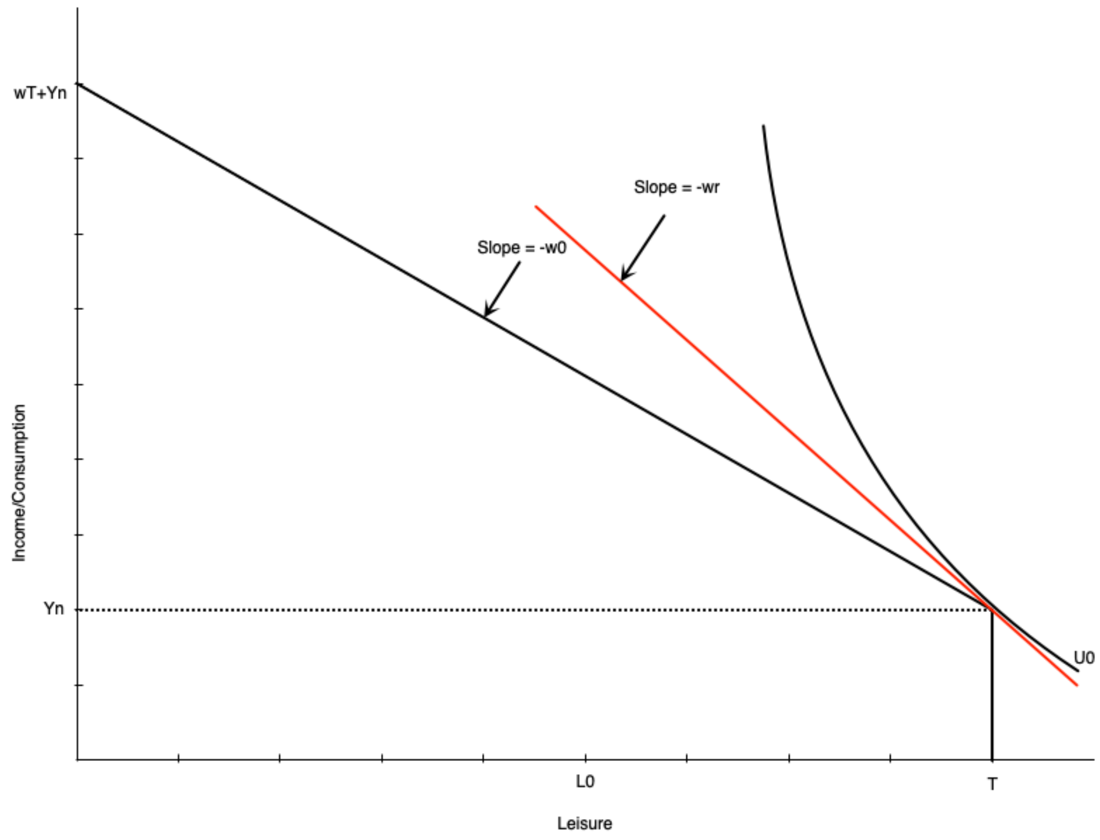
- Graph shows an optimum with participation
- Optimum is where indifference curve is tangent to budget constraint
- At the optimum, MRS = wage rate
  - Willingness to trade leisure for consumption (MRS) equals ability to trade them (wage)
- Optimal work hours are $h_0 = T - L_0$
- Optimal consumption is $C_0 = W_0 h_0 + Y_n$

Consumer Optimum



- The reservation wage is the lowest wage at which a person will choose to work
- It occurs where the budget constraint is tangent to the indifference curve at full leisure
    - At that point, MRS = reservation wage
- If the wage is any lower, the person will choose not to work
- If it is any higher, they will work positive hours
- The red line shows the hypothetical reservation wage
    - In this case, it is w_r

Consumer Optimum



- Graph to the left shows a non-participation optimum
- The slope of the indifference curve is steeper than the budget constraint at full leisure
  - The extra consumption you want from one less hour of leisure (MRS) is more than you can get from working an hour (wage)
- So your optimal choice is to not work at all
- Indifference curve touches the budget constraint at the kink
- Notice the wage w_0 is lower than the reservation wage w_r

Comparative Statics

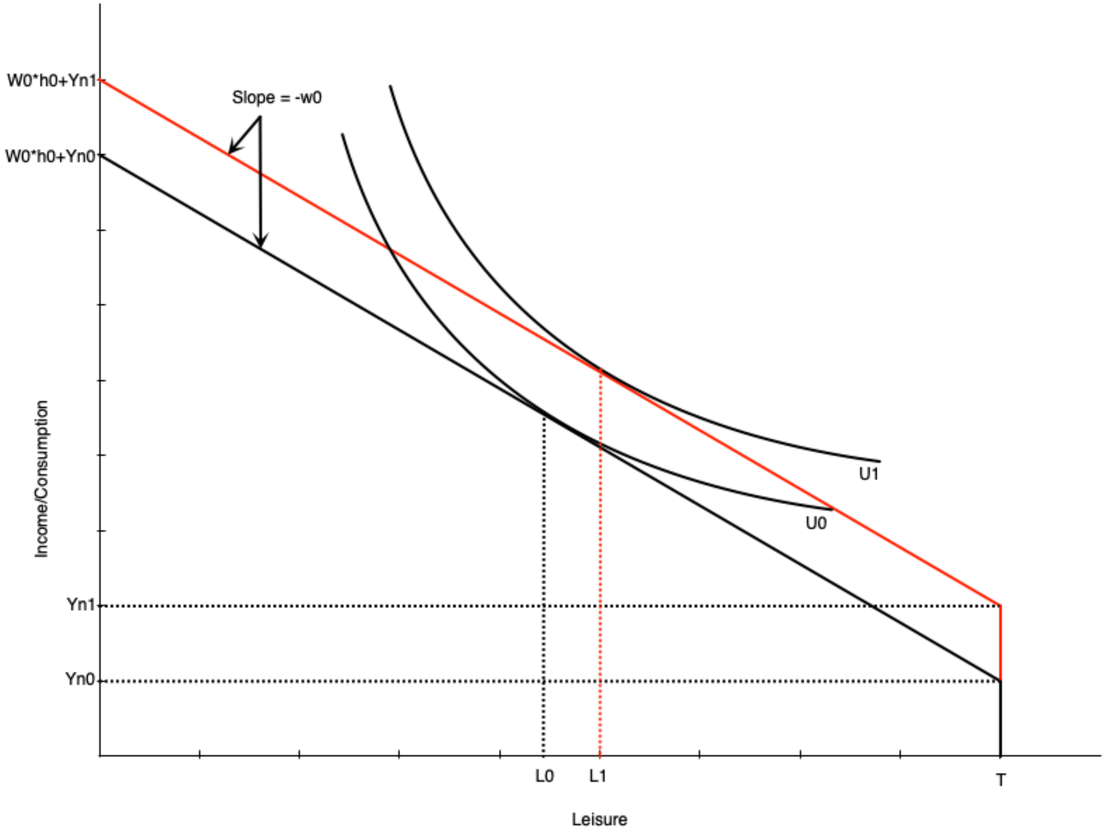
Introduction

- The model can be used to predict how changes in the environment affect work decisions
- Changes could be to the wage or non-labour income
- Often these arise because of policy changes
  - Introduction or increase in welfare payment
  - Wage subsidy
  - Tax credits
  - Employment insurance
  - Child care
- We can examine these change with our model

Increase in Non-Labour Income

- Suppose that individuals receive an increase in non-labour income
  - Could be from a government transfer program
- How does this affect optimal work hours?
- Depends on whether leisure is a normal or inferior good
  - Normal good: leisure increases when non-labour income increases
  - Inferior good: leisure decreases when non-labour income increases
- Generally economists think of leisure as normal

Increase in Non-Labour Income

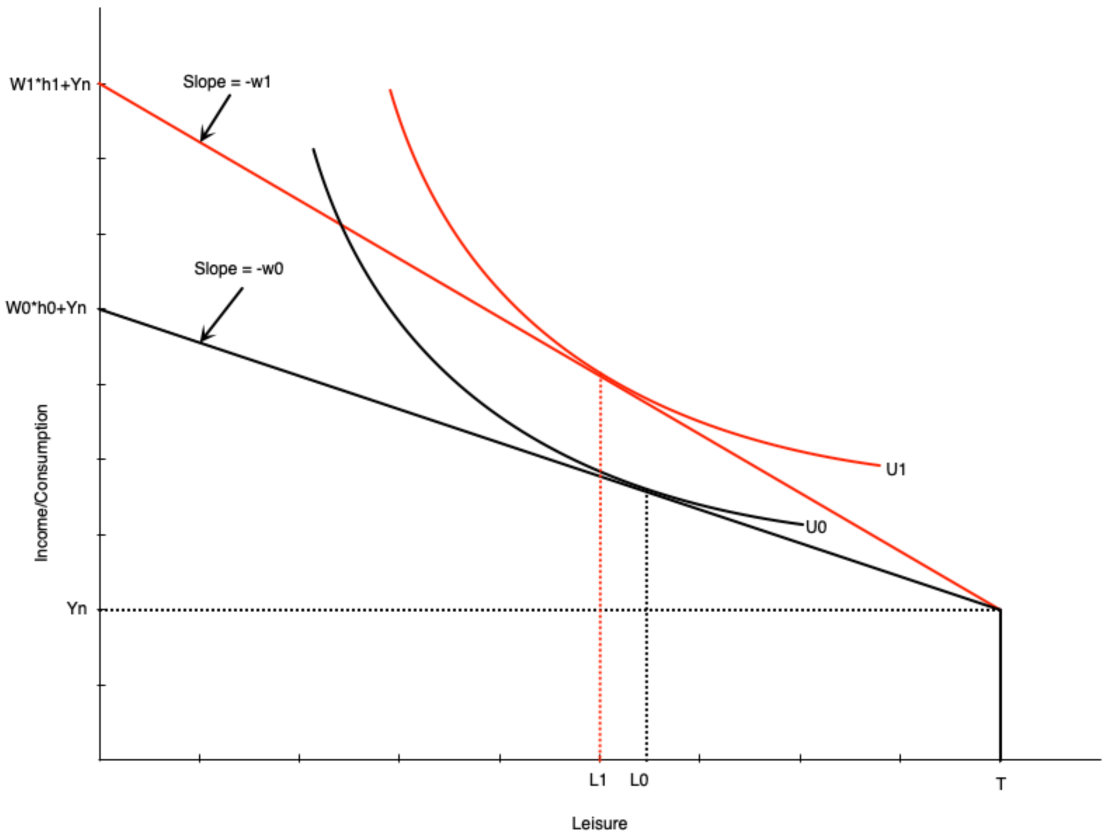


- Graph shows an increase in non-labour income from Y_{n0} to Y_{n1}
- Budget line shifts up vertically but parallel
  - No change in slope because wage is the same

-If leisure is a normal good, optimal leisure increases from L_0 to L_1

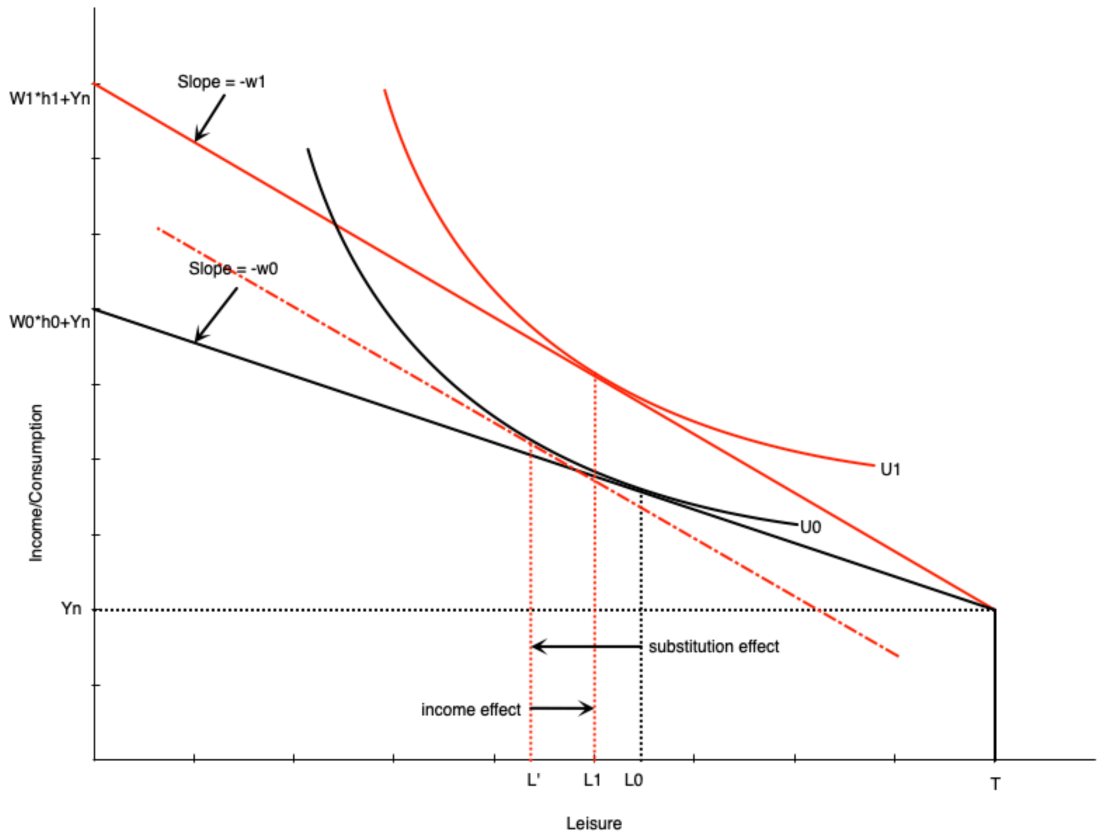
- Means work hours decrease by the same amount
- Consumption also increases
- This illustrates a pure **income effect**
  - Increases in income increase demand for normal goods

Increase in the Wage



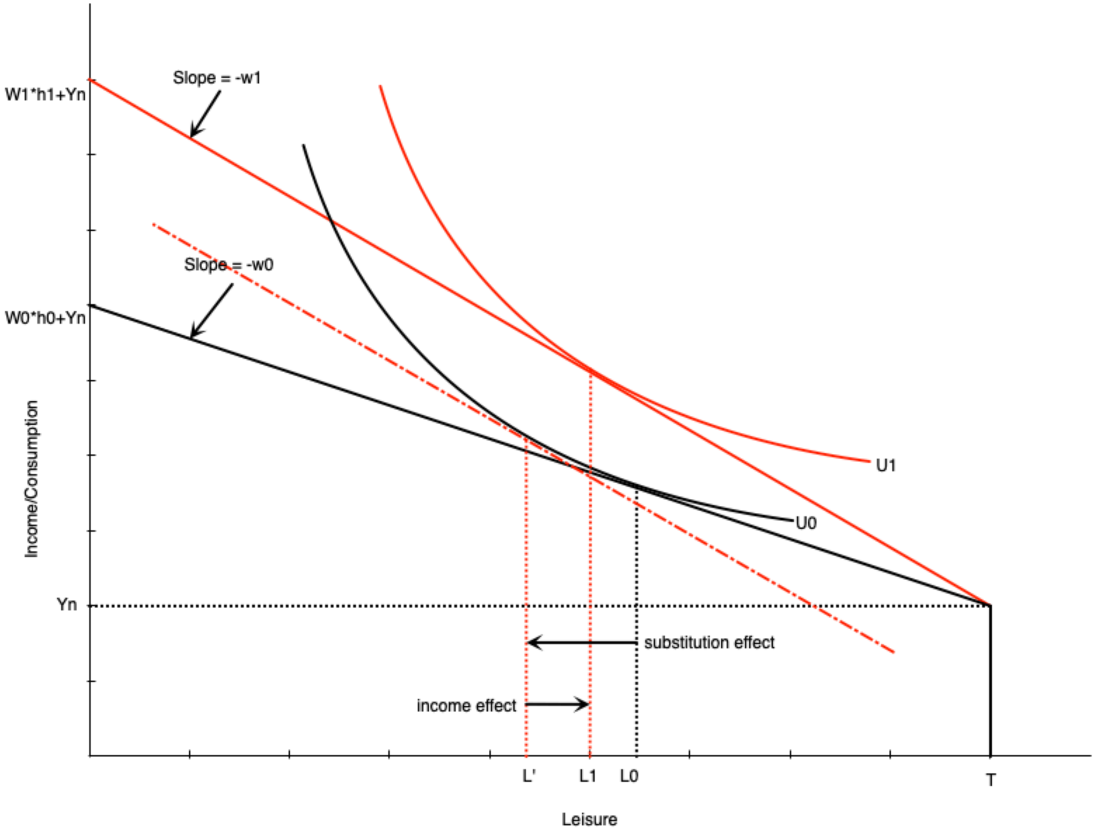
- If the wage changes, the budget constraint pivots
- In graph to the left, the wage increases from w_0 to w_1
- The slope becomes steeper
- Full income also increases because working all hours earns more
- Graph is drawn such that leisure falls from L_0 to L_1 at the new wage rate
- There are two effects underlying this change
  - Substitution effect
  - Income effect

Increase in the Wage



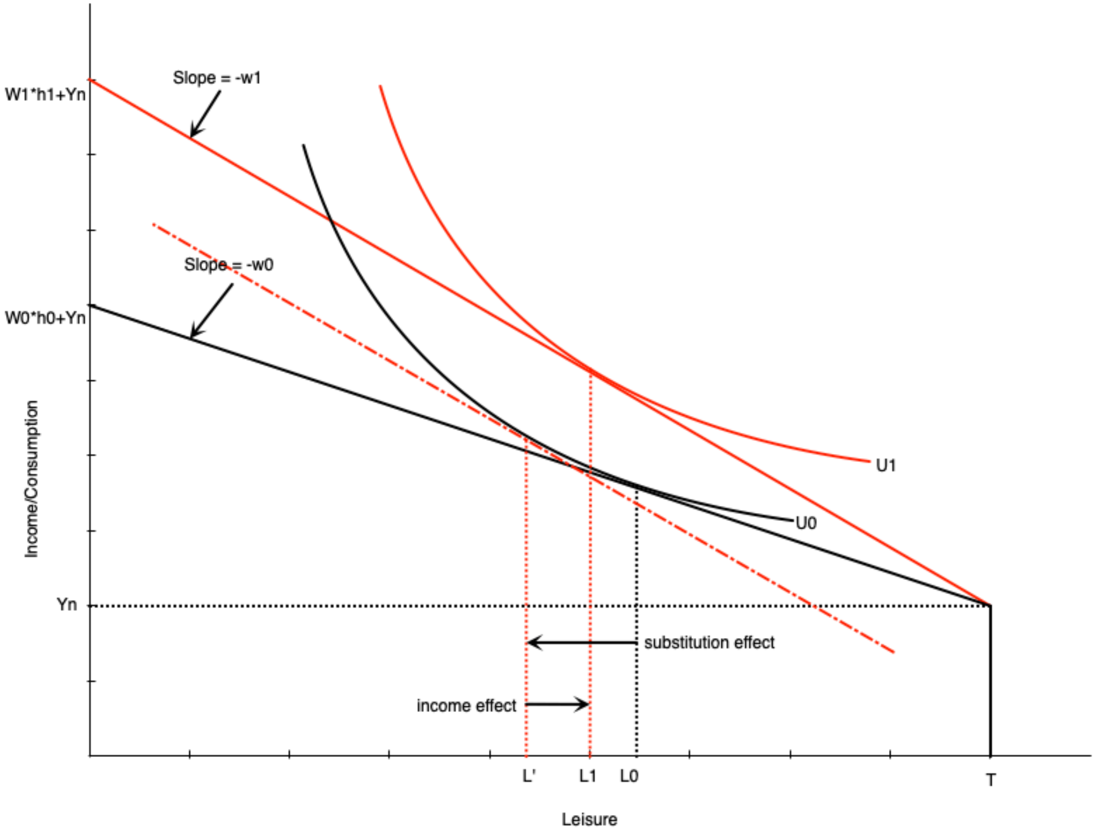
- **Substitution Effect**: change in leisure when wage changes but utility is held constant
  - When wage increases, leisure becomes more expensive
  - People substitute away from leisure and towards consumption
  - To measure this, draw a hypothetical budget line with the new slope but tangent to the original indifference curve
  - Difference in leisure measures substitution effect

Increase in the Wage



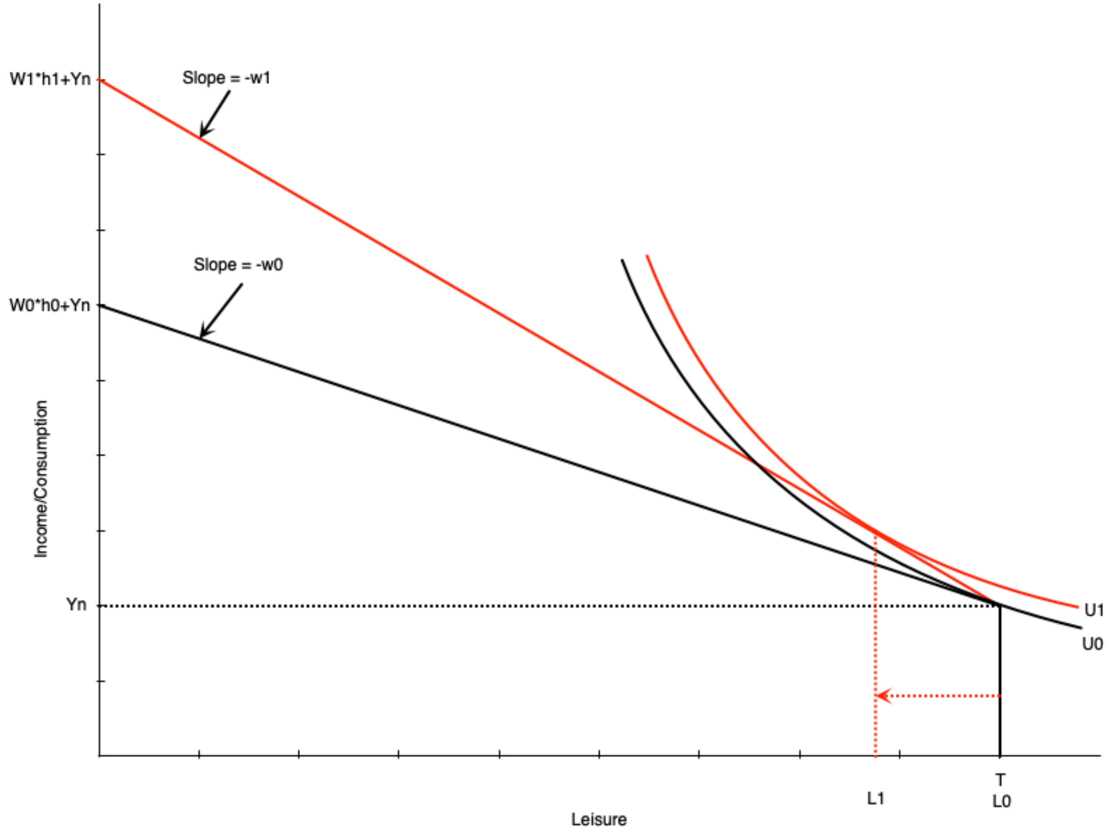
- **Income Effect**: change in leisure when income changes but wage is held constant
  - When wage increases, workers become richer at same hours of work
  - Want to consume more of all normal goods
  - To measure this, move from hypothetical budget line to new budget line
  - Difference in leisure measures income effect

Increase in the Wage



- The income and substitution effects move in opposite directions
- For a wage increase
  - Substitution effect encourages more work (less leisure)
  - Income effect encourages less work (more leisure)
- The overall effect on work hours depends on which effect is stronger
- Graph on the left shows a stronger substitution effect

Increase in the Wage

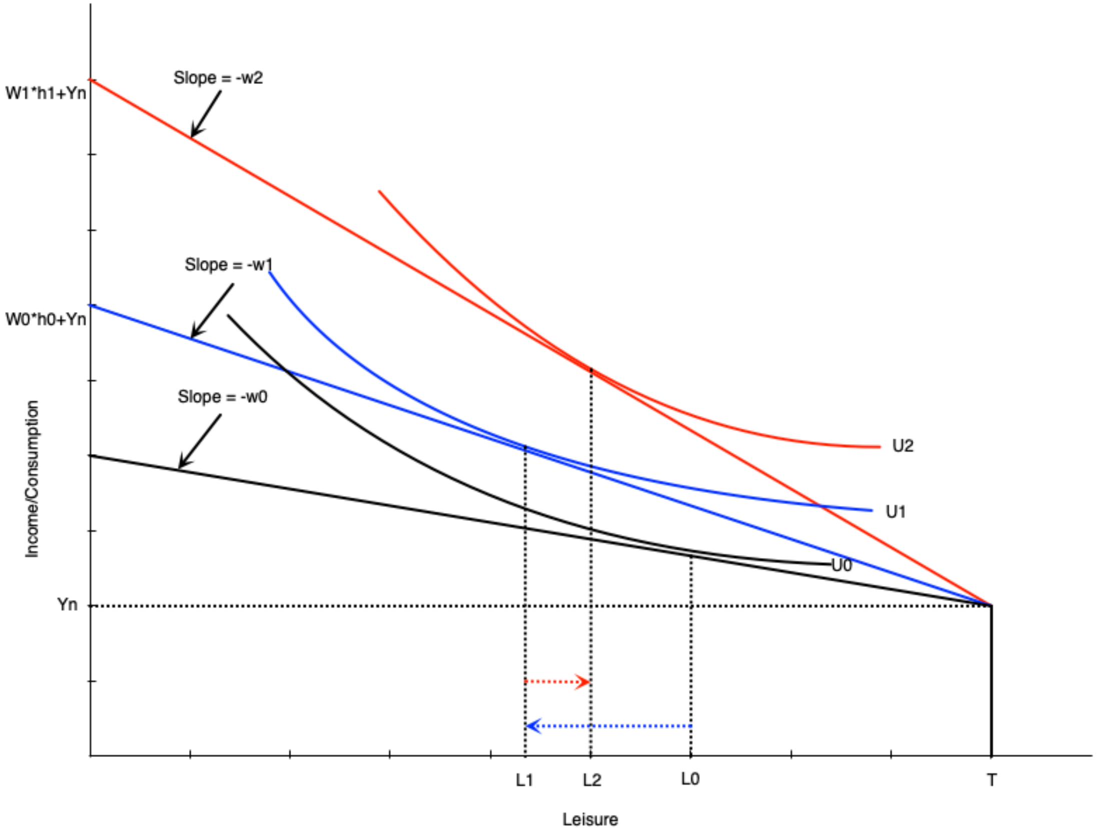


- Changing the wage can also affect participation
- Recall that workers do not work if the wage is below their reservation wage
- If an increase in the wage moves it above the reservation wage, they will start working
- Graph to the right shows how this would happen

The Individual Labour Supply Curve

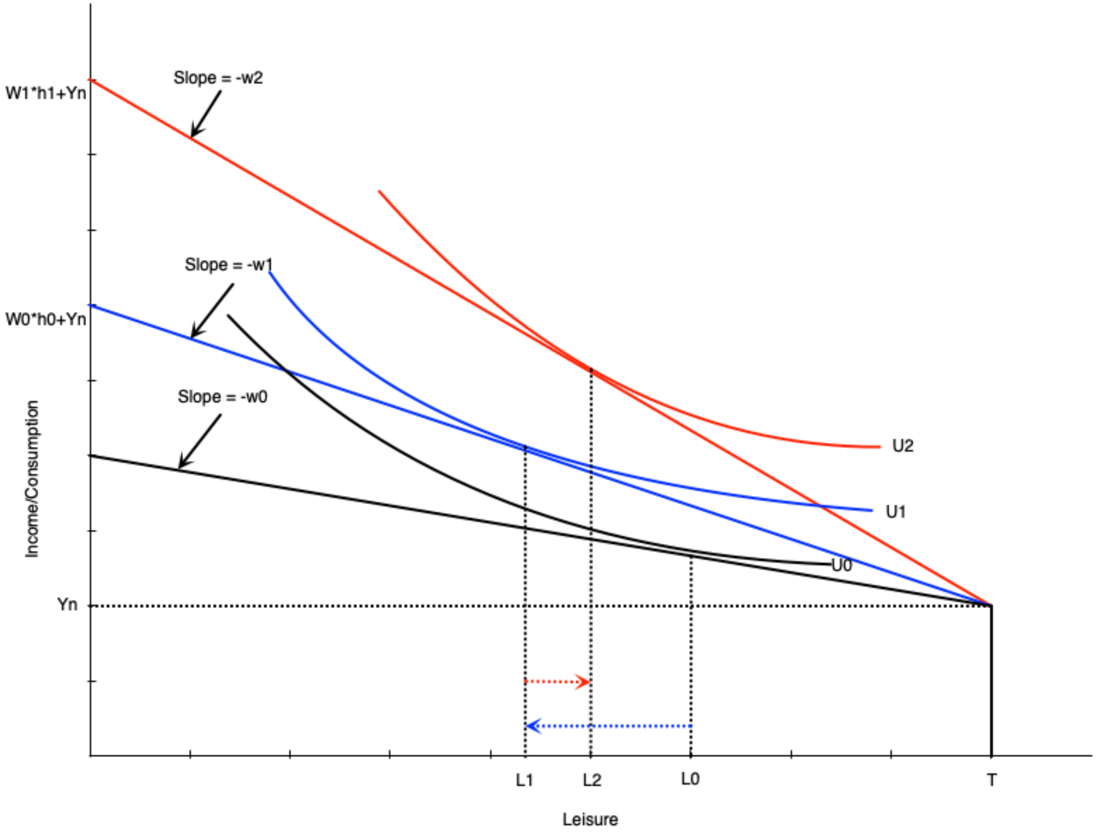
- The labour supply curve shows the relationship between the wage rate and hours worked
- It is derived from the consumer optimum at different wage rates
- We will see that the labour supply curve can be backward bending
  - It is upward sloping when the substitution effect is stronger
  - It is downward sloping when the income effect is stronger
  - The curve tends to bend backwards at higher wage rates

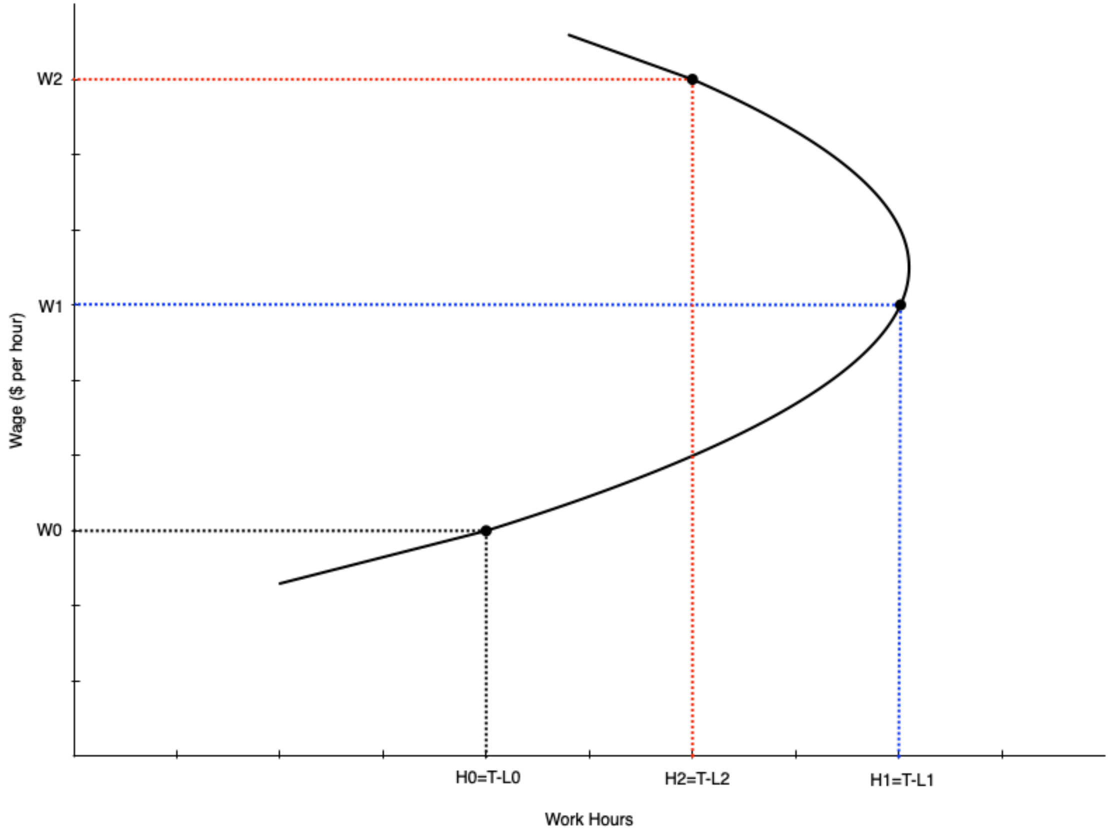
The Individual Labour Supply Curve



- Suppose wage goes from w_0 to w_1 to w_2
- At w_0, work hours are $h_0 = T - L_0$
- At w_1, work hours increase to $h_1 = T - L_1$
  - substitution effect stronger
- At w_2, work hours decrease to $h_2 = T - L_2$
  - income effect stronger

The Individual Labour Supply Curve





Summary

- We have used the labour supply model to examine changes in the wage
- For people participating, wage changes alter leisure and work hours
- If the wage goes up then
  - Work hours go up if the substition effect is strongest
  - Work hours go down if the income effect is strongest
- For non-participants, a wage increase can induce them to enter the labour force
  - If the wage goes above their reservation wage
- The labour supply curve maps different wages to work hours
- It slopes up at lower wages, but can bend backwards at higher wages

Empirical Evidence

Introduction

- There are many labour economists working on evidence of these theories
- We will look at a few examples of empirical evidence on labour supply
- First we look at evidence on participation
  - The **extensive margin** of labour supply
  - Many studies look specifically at married women
- Then at evidence on work hours
  - The **intensive margin** of labour supply

Participation

- Table below shows participation rates for married women
- Broken down by age, children, education, spousal income
- Each of these demographic factors are related to wage
  - e.g. higher education -> higher wage
- We expect to see that where wage is higher, participation is also higher
- This is not a perfect way to examine this relationship
  - There could be other related factors that drive participation

Participation

TABLE 2.2 Labour Force Participation Rates of Married Women, Canada 2017				
	Percentage of Married Women	Participation Rate[a]	Raw Difference from Base Group[b]	Adjusted Difference from Base Group[c]
All Women, Total		**75.3**		
Age				
20–24	1.1	89.4	N/A	N/A
25–34	18.9	84.4	–5.0	–3.7
35–44	23.9	85.5	–3.9	–3.0
45–54	24.0	84.8	–4.6	–7.3
55–64	23.8	64.8	–24.6	–22.3
65–69	8.4	27.5	–61.9	–52.6
Age of youngest child, under 18 years old, at home				
No children under 18	60.1	70.9	N/A	N/A
0–5 years old	18.4	77.4	6.5	–13.9
6–9 years old	7.7	84.6	13.7	–5.4
10–15 years old	10.1	87.0	16.1	–0.7
16–17 years old	3.7	86.3	15.4	0.2
Education				
Less than high school graduation	8.2	49.6	N/A	N/A
High school graduate or some post-secondary	23.6	67.9	18.3	12.1
Non-university, post-secondary certificate	30.7	77.8	28.2	18.8
University degree	37.6	83.7	34.1	22.4
Spouse's income				
None (or negative)	18.0	49.7	N/A	N/A
$1–$9,999	8.7	62.0	12.3	7.7
$10,000–$24,999	9.1	78.6	28.9	17.5
$25,000–$49,999	18.2	82.3	32.6	18.4
$50,000–$74,999	17.1	87.1	37.4	21.0
$75,000–$99,999	12.4	86.3	36.6	18.9
$100,000–$149,999	10.7	84.2	34.5	17.1
Over $150,000	5.7	74.1	24.4	7.4

Elasticity of Labour Supply

- The **elasticity of labour supply** is the percentage change in labour supply from a 1% change in the wage
- Measures how responsive work hours are to changes in the wage
- There are different ways of measuring elasticity
  - **Uncompensated Elasticity**: measures total change in work hours from a wage change
    - Includes both income and substitution effects
    - Effect is theoretically ambiguous
  - **Compensated Elasticity**: measures change in work hours after removing income effect
    - Isolates the substitution effect
    - Should be positive

Elasticity of Labour Supply

TABLE 2.3 Estimated Elasticities of Labour Supply

Sex	Uncompensated (Gross, Total) Wage Elasticity of Supply	Compensated (Substitution) Wage Elasticity of Supply	Income Elasticity of Supply
Both sexes	0.0 to 0.2	0.1 to 0.3	–0.1 to 0.0
Men and single women	–0.1 to 0.2	0.1 to 0.3	–0.1 to 0.0
Married women	0.1 to 0.3	0.2 to 0.4	–0.1 to 0.0

NOTES: The uncompensated elasticity represents the elasticity for the choice of hours to work, conditional on working.

SOURCE: Adapted from Table 2 of McClelland and Mok (2012), in their review of U.S. studies for the Congressional Budget Office.

Elasticity of Labour Supply

- Table above shows summary of uncompensated and compensated elasticities from different studies
- Separates by gender
  - We might expect men and women to respond differently to wage changes
  - Women potentially have more flexibility in work hours and respond more
- Table shows uniformly *inelastic* labour supply
  - Elasticity less than 1 in absolute value
  - A 1% increase in wage leads to roughly 0.1% to 0.3% change in work hours
- Compensated elasticity is positive
  - Women respond more to wage changes
- Income effect is generally negative or small

Applications

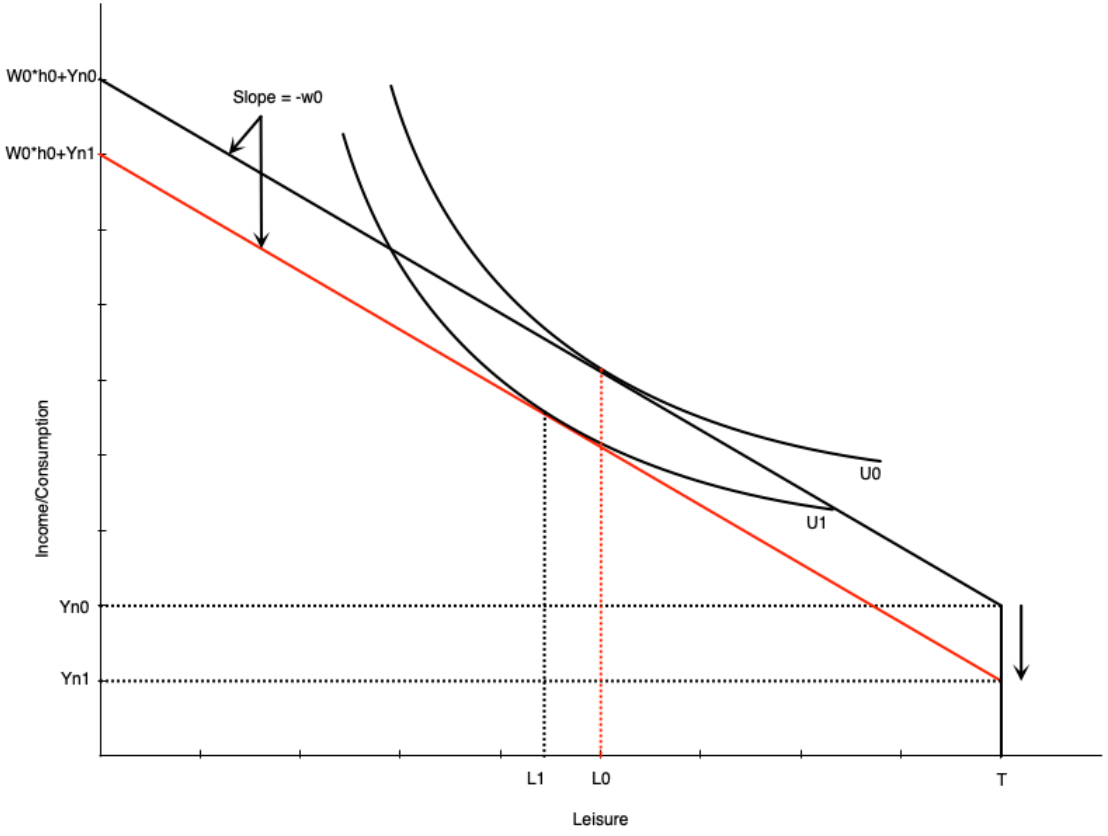
Introduction

- We can use the model to analyze various situations faced by workers
- In this section we will look at
  - Changes in family income (added worker effects)
  - Discouraged workers
  - Moonlighting (holding multiple jobs)
  - Overtime
  - Flex work
- The list is not exhaustive
- In the next chapter we will look at government policies

Added Worker Effect

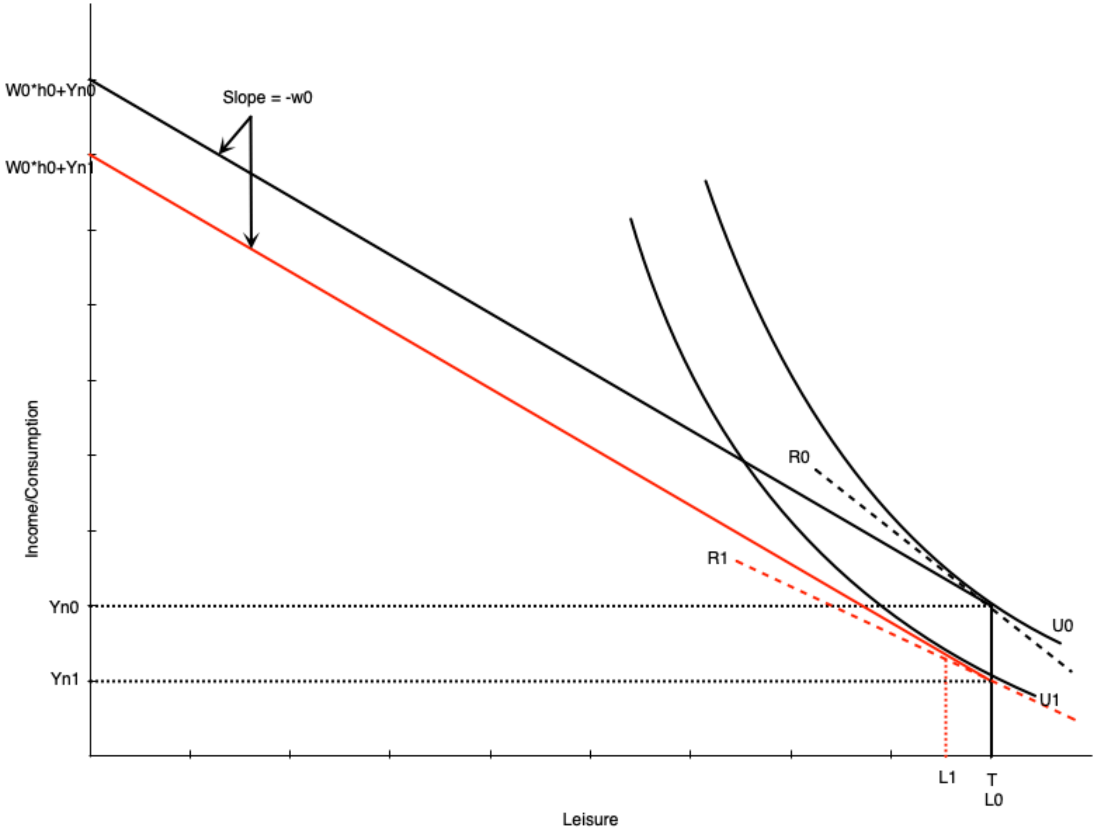
- The **added worker effect** occurs when a negative income shock to a family leads to increased labour supply from secondary earners
  - e.g. husband loses job, wife enters labour force or increases work hours
- Explicitly considers the family unit rather than treating labour supply as purely individual
- Spousal income is treated as a person's non-labour income
- In the model we can look at what happens when this amount falls

Added Worker Effect



- When non-labour income falls, the budget constraint shifts down
- If leisure is a normal good, optimal leisure falls from L_0 to L_1
- Work hours increase from h_0 to h_1
- Means that the spouse increases work hours to compensate for lost family income

Added Worker Effect

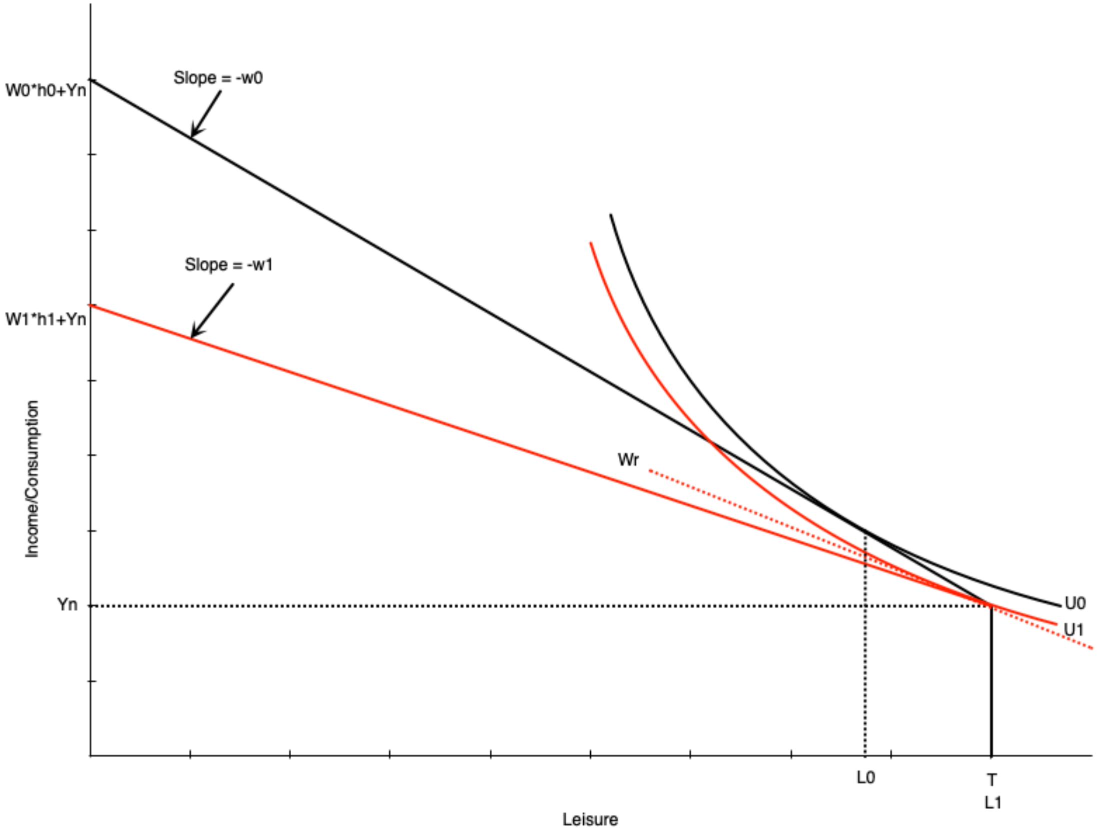


- If the spouse was not already working, the fall in income can induce participation
- Happens because the fall in income reduces the reservation wage
  - If the wage is now above the reservation wage, they will start working
- Graph to the left shows how this happens
  - The slope if reservation wage R_1 is smaller than R_0 after the drop in non-laboir income
  - Person enters the labour market in response because wage is now above reservation wage

Hidden Unemployment

- We discussed a problem with the unemployment rate in that it does not include discouraged workers
- **Discouraged workers**: people who could work but have given up looking and dropped out of the labour force
- Our model can explain this behaviour
- In bad times, the wage falls
- If the wage falls below a person's reservation wage, they will drop out of the labour force

Hidden Uneployment

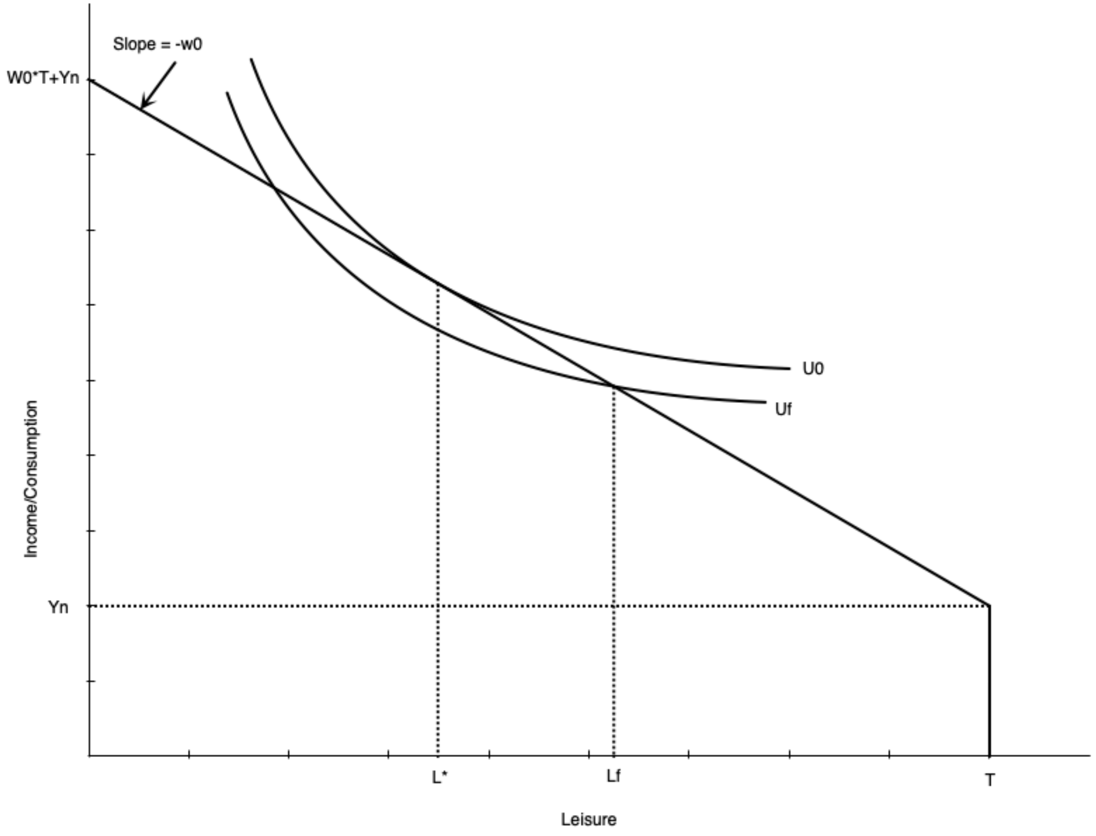


- Graph shows worker initially working positive hours
- Wage falls from w_0 to w_1
- w_1 is below the reservation wage w_r
  - The person will drop out of the labour force
- Now use all their time for leisure
  - Consume non-labour income Y_n

Moonlighting

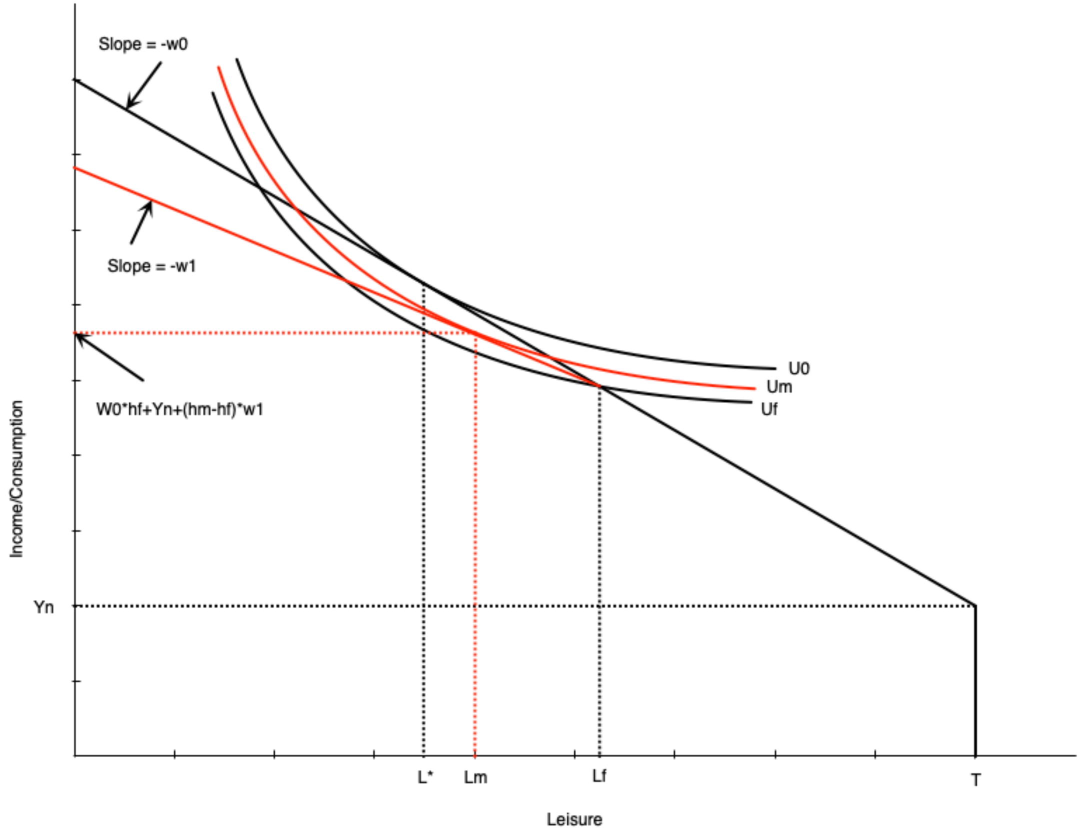
- **Moonlighting**: working additional jobs at night or on weekends
  - Bit of an old term where main jobs were generally 9-5
  - Extra work occured after that
  - In 2025 secondary jobs can happen at any time
- Happens when people have a fixed hours constraint
  - Example: working full time, but wishing to work longer
  - People with fixed hours can be **underemployed**
- People may take a second job, possibly at a lower wage

Moonlighting



- In this graph people ideally would work $h^* = T - L^*$ hours
  - Maximizes utility subject to budget constraint
- Can only work $h_f = T - L_f$ hours at their main job
  - Employer does not allow more hours
- This puts them on a lower indifference curve U_f
- The outcome is not optimal

Moonlighting

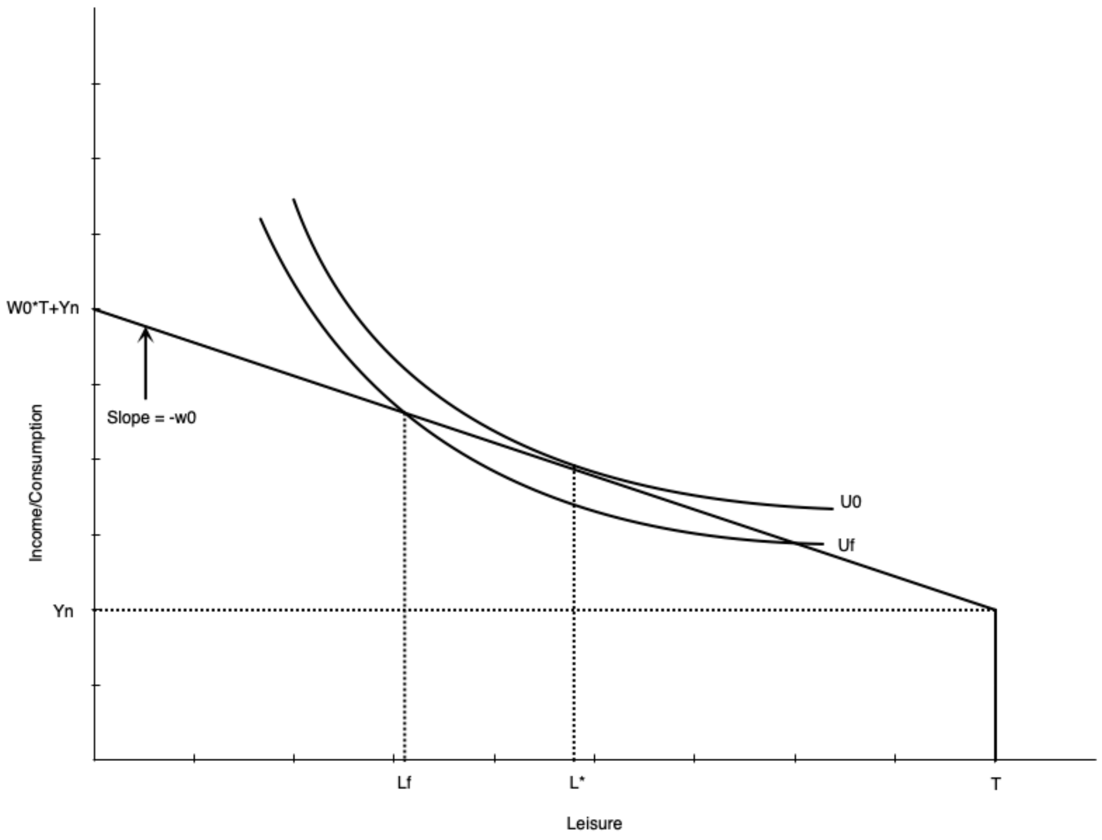


- Now imagine they can take a second job at a lower wage rate
- The budget constraint pivots out from the fixed hours point
- Beyond those hours, the wage falls
- Allows worker to access higher utility curve
  - They can increase hours worked
  - But not as much as if they could extend hours in main job

Overtime

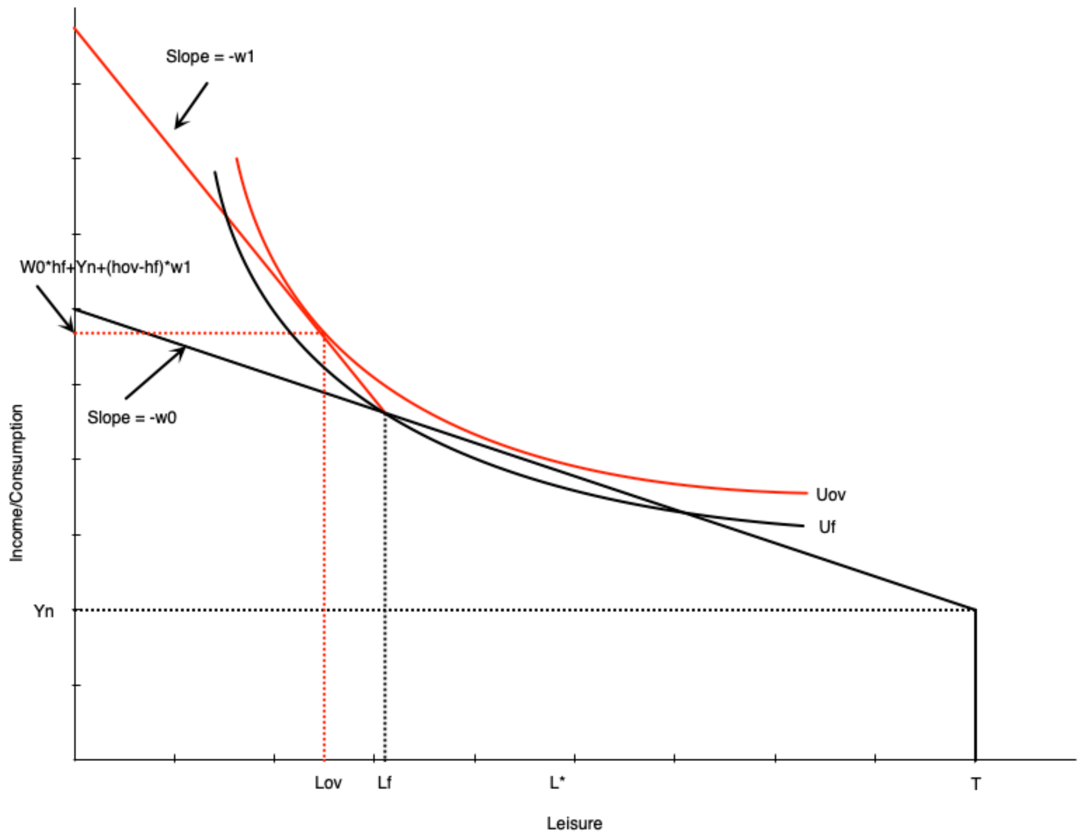
- Sometimes people are working too much at the main job
- Also happens with a fixed hours constraint
  - People want to work less at given wage
  - Employer does not allow less hours
  - They are **overemployed**
- At given wage they cannot maximize utility subject to constraints

Overtime



- In this graph, people would ideally work $h^* = T - L^*$ hours
  - Maximizes utility subject to budget constraint
- But employer requires them to work $h_f = T - L_f$ hours
  - Employer does not allow fewer hours
- Puts them on a lower indifference curve U_f
- The outcome is not optimal

Overtime

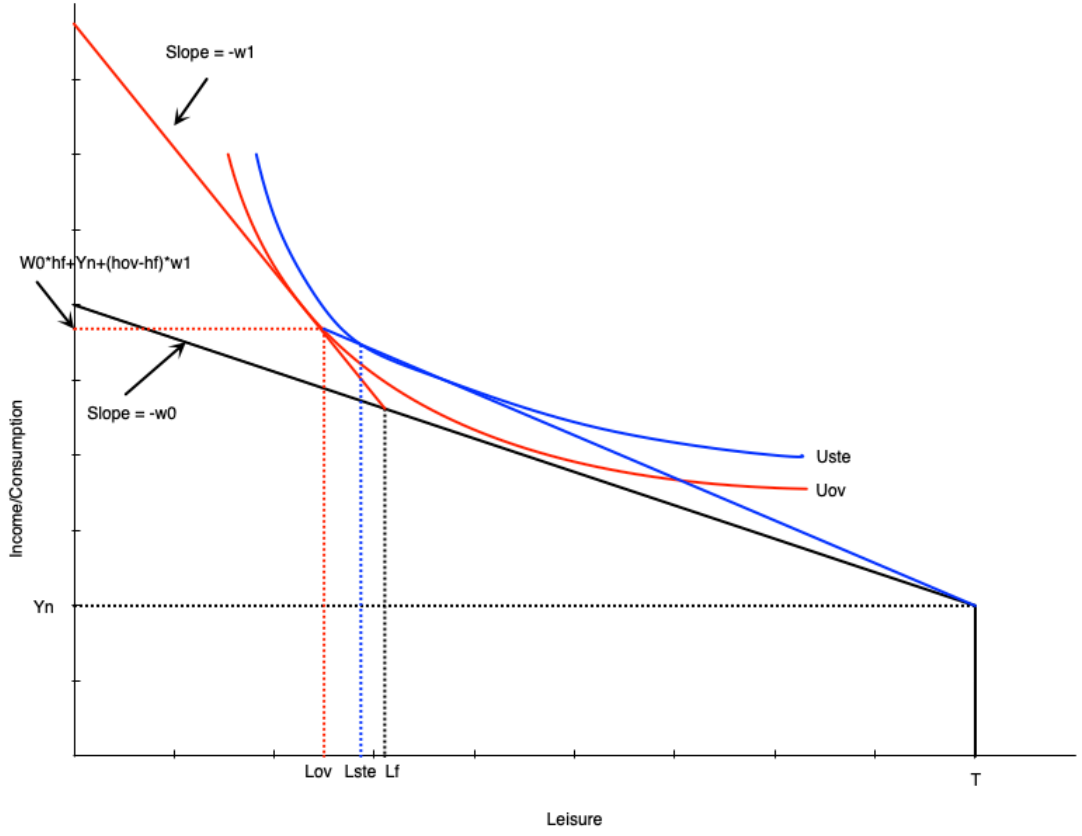


- If the employer offers overtime wages, the budget constraint pivots out from the fixed hours point
  - Beyond those hours, the wage increases from w_0 to w_1
- Allows worker to access higher utility curve
- Also increases hours worked even though they are initially overemployed
  - The higher wage induces a substitution effect to work more
  - There is a small income effect that only applies to the extra hours

Overtime

- An alternative to overtime is for the firm to just pay more for all hours
  - Makes sense for a firm if they are constantly paying overtime
- That does not generate the same optimal work hours as overtime pay
  - Income effect is larger with straight time equivalent
  - Becauses wage increase applies to *all* hours worked
  - In overtime, income effect only applies to extra hours worked
- Worker chooses to less with straight time equivalent pay

Overtime



- Straight-time equivalent work is depicted to the left
- The blue line goes from the fixed hours point with slope w_1
  - Generates the equivalent income as overtime pay at the optimum
- Worker chooses fewer work hours than overtime
  - Due to income effect of having higher wage at all work hours

Summary

Summary

- We have explored the labour-leisure model for labour supply
- The model is based on utility maximization subject to a budget constraint
- The model can explain both participation and work hours decisions
- Changes in non-labour income and the wage affect work decisions
- The labour supply curve can be backward bending
- The model can be used to analyze various situations faced by workers
  - Added worker effects
  - Discouraged workers
  - Moonlighting
  - Overtime
- Next chapter we will analyze government policy within this model